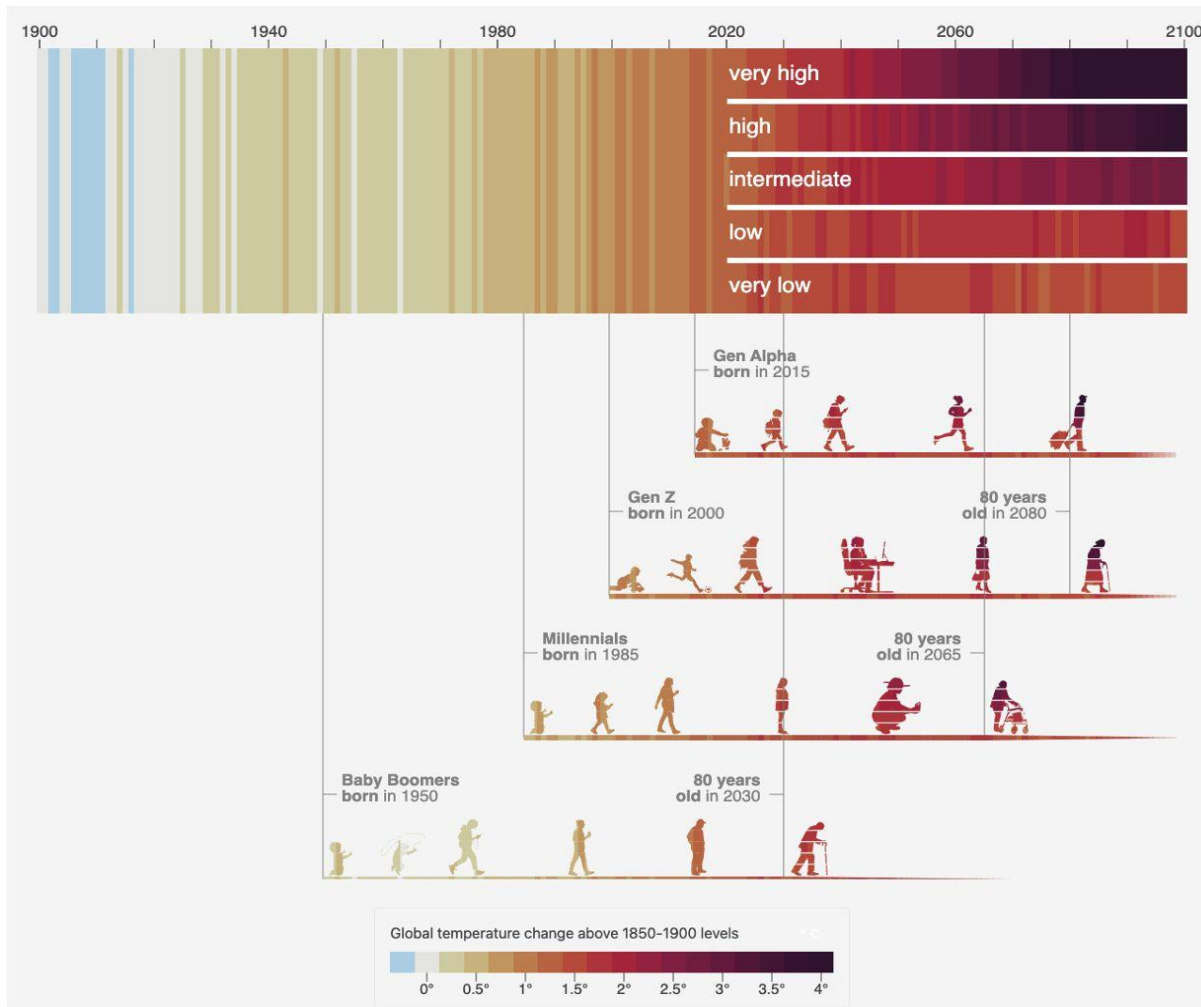


Climate Change Governance: Science, Data & Models



Lecture 10: Energy System Transitions

ENGG562/ENV562 – Fall 2026

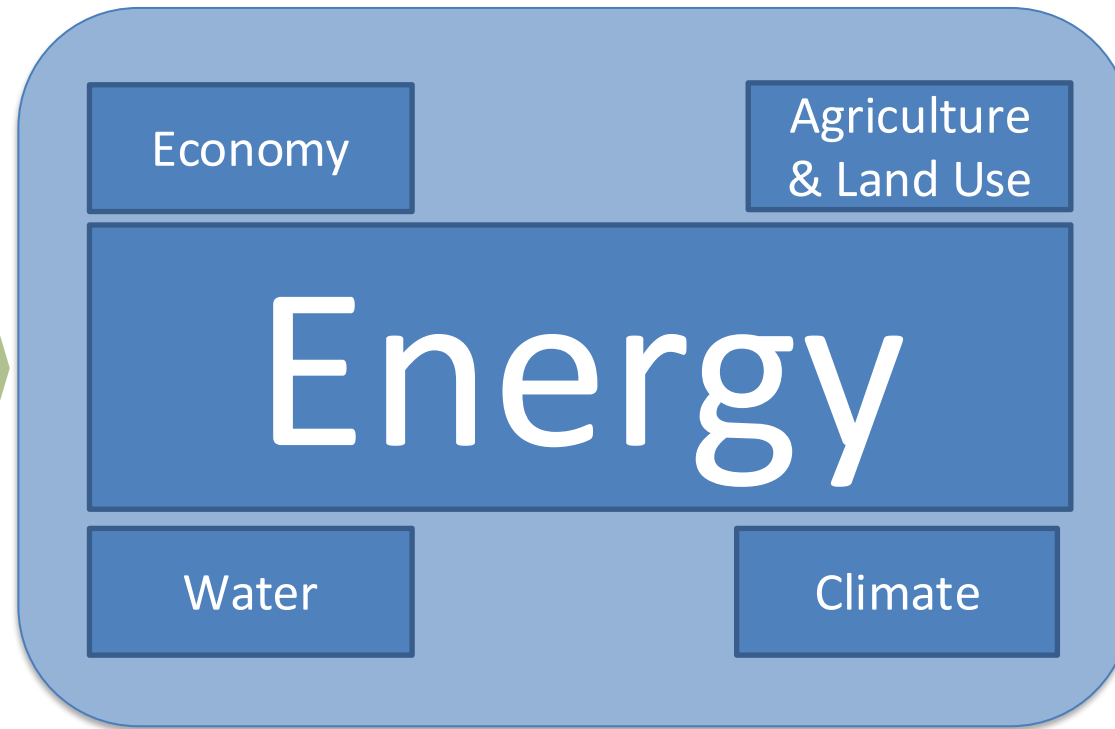
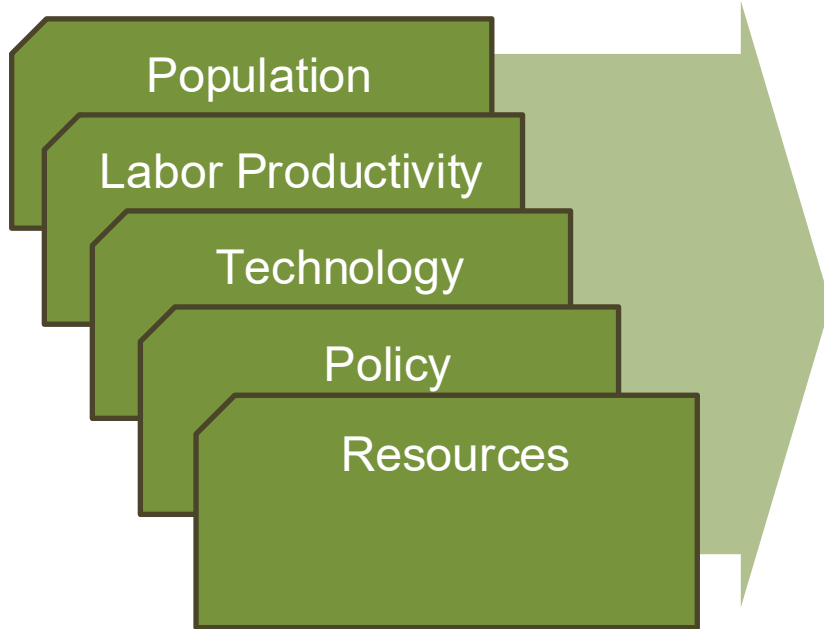
Muhammad Awais

Agenda for today

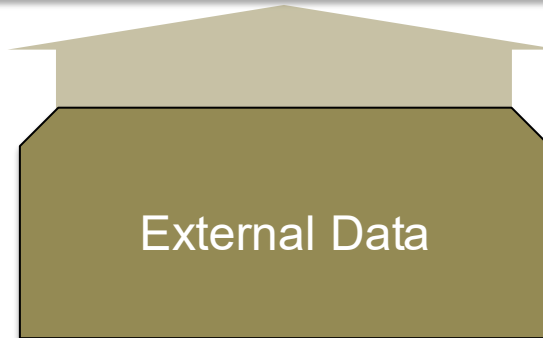
- Importance of the energy system in integrated assessment modeling
- Components of the energy system
- Quick refreshers
 - Energy Units
 - International climate policy
 - Energy technologies
- Key characteristics of long-term energy system transformations
- Modeling of non-economic factors
- Cutting-edge research in modeling of energy systems in IAMs

The energy system is at the heart of integrated assessment models...

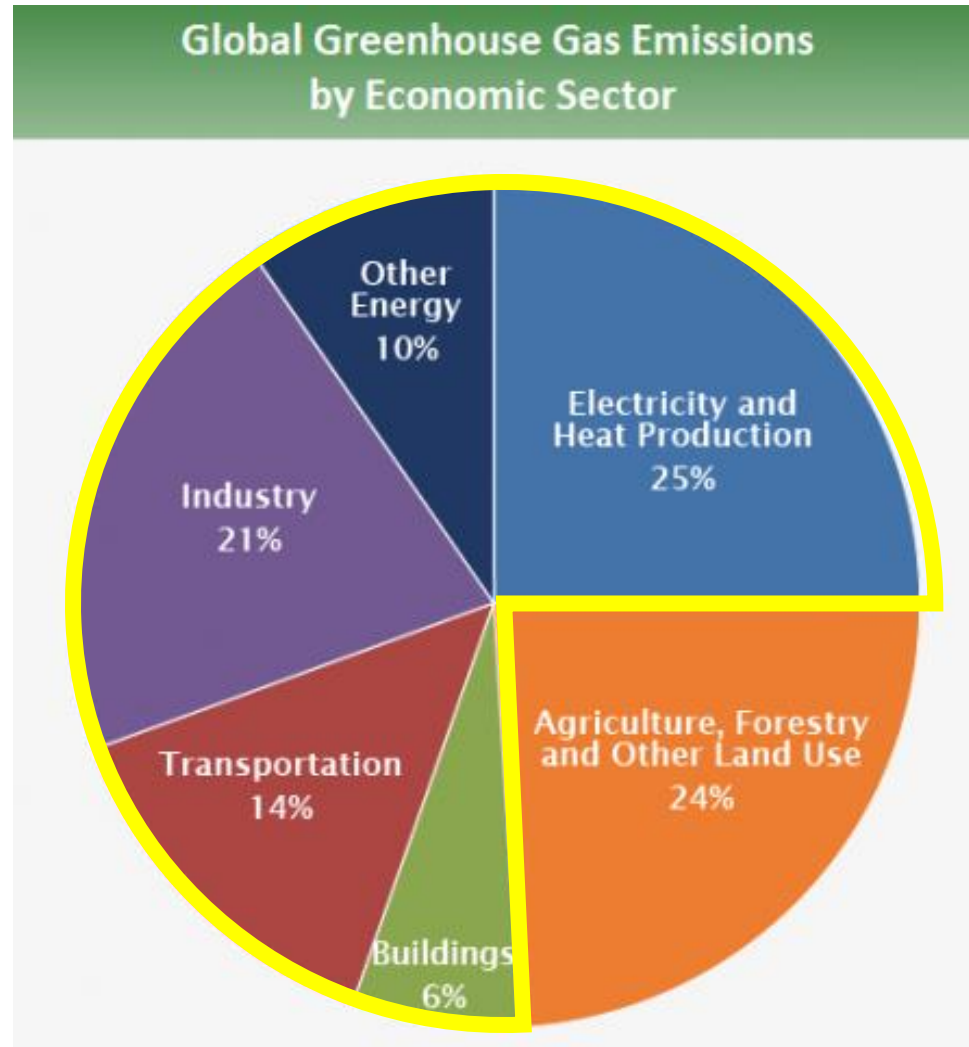
Exogenous Assumptions



Outputs of IAMs



...because the energy system is at the heart of the climate problem...



Source: <https://www.epa.gov/ghgemissions/global-greenhouse-gas-emissions-data>

Thinking about the unseen pillars

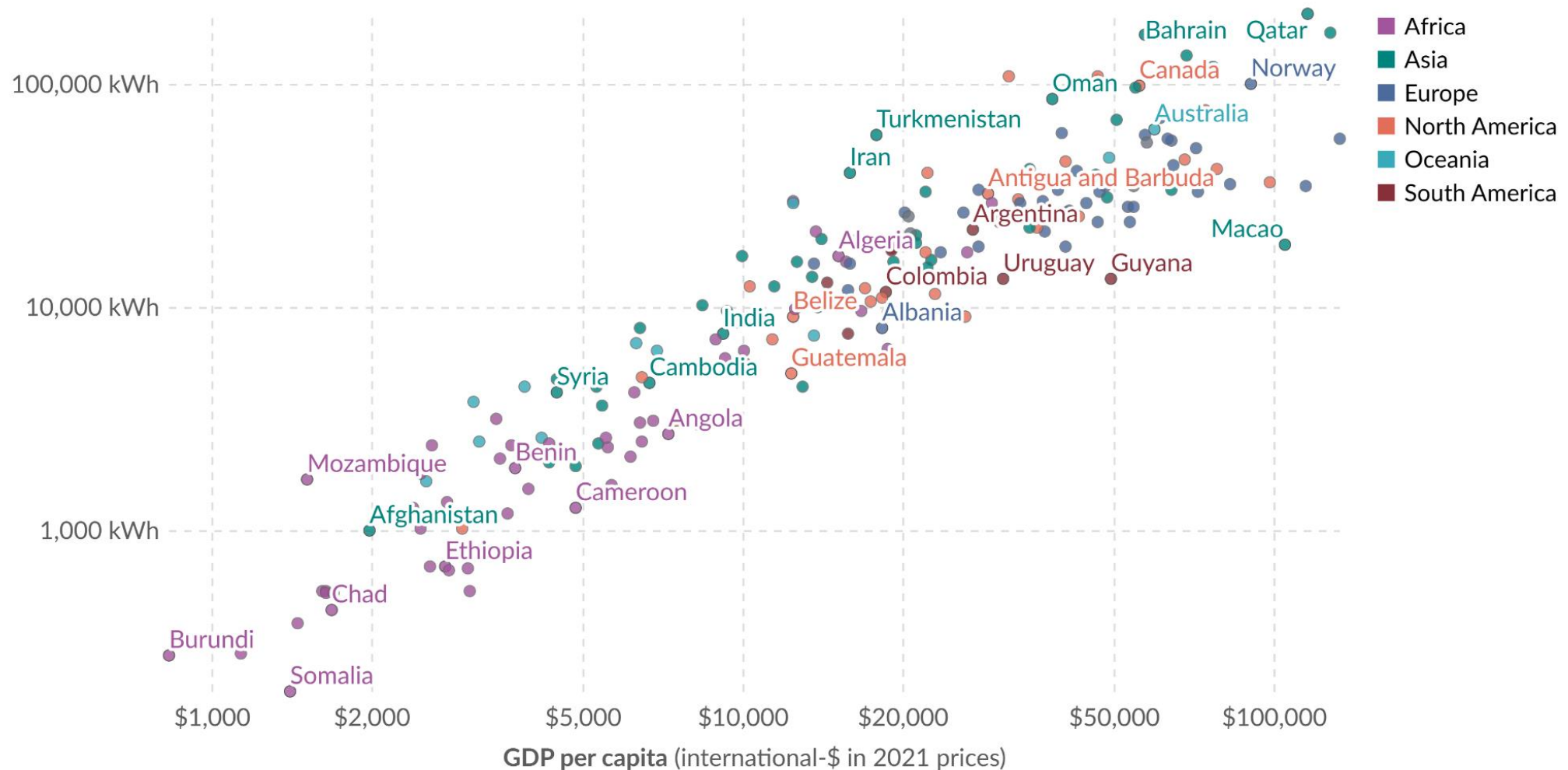
What does it take to make
cities grow and citizens
prosperous?



Energy use per person vs. GDP per capita, 2023

Energy refers to primary energy¹, measured in kilowatt-hours² per person, using the substitution method³. Gross domestic product (GDP) is adjusted for inflation and differences in living costs between countries.

Per capita energy consumption (kilowatt-hours)



Data source: U.S. Energy Information Administration (2025) and other sources

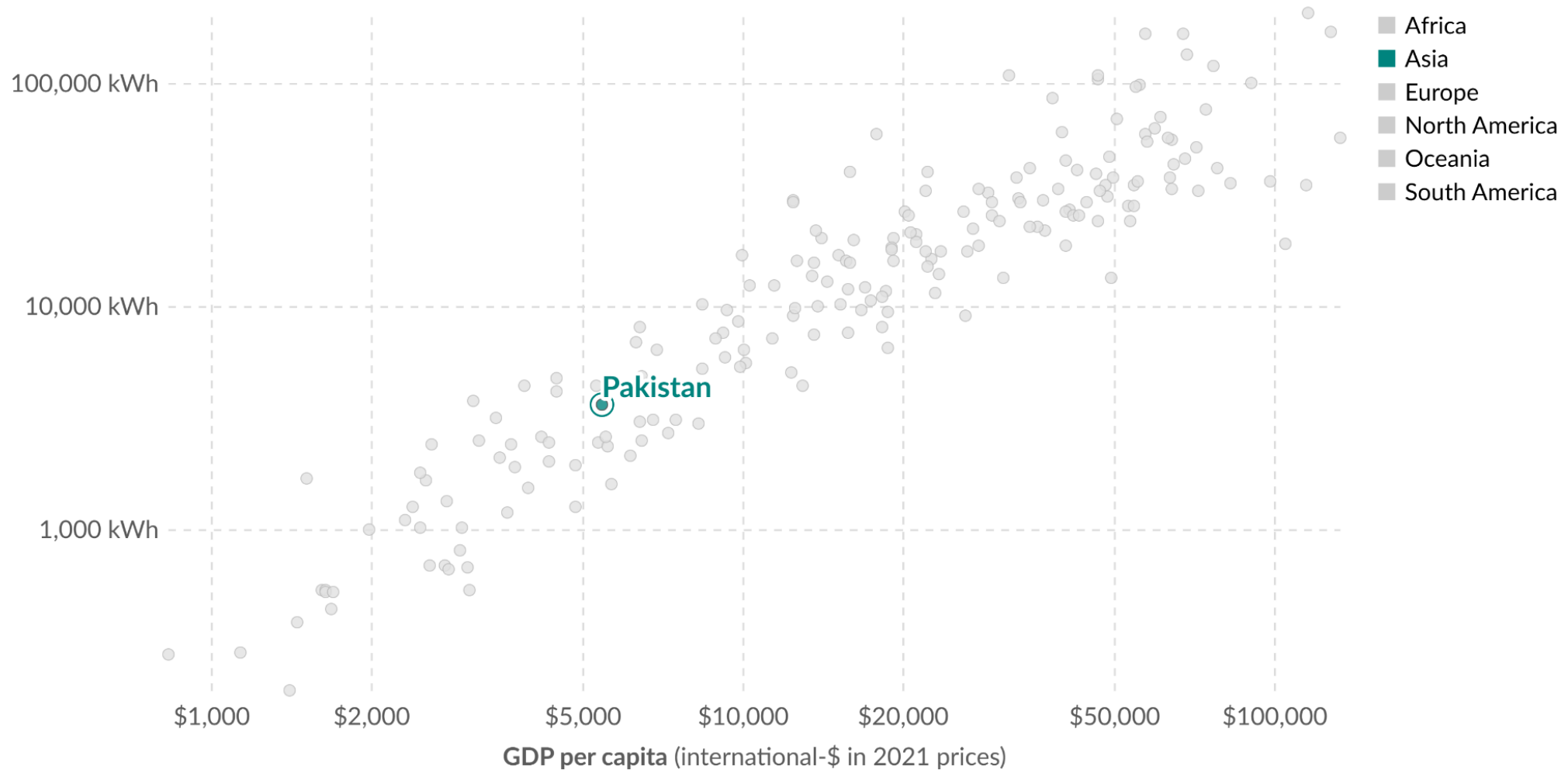
OurWorldinData.org/energy | CC BY

Note: GDP data is expressed in international-\$⁴ at 2021 prices.

Energy use per person vs. GDP per capita, 2023

Energy refers to primary energy¹, measured in kilowatt-hours² per person, using the substitution method³. Gross domestic product (GDP) is adjusted for inflation and differences in living costs between countries.

Per capita energy consumption (kilowatt-hours)



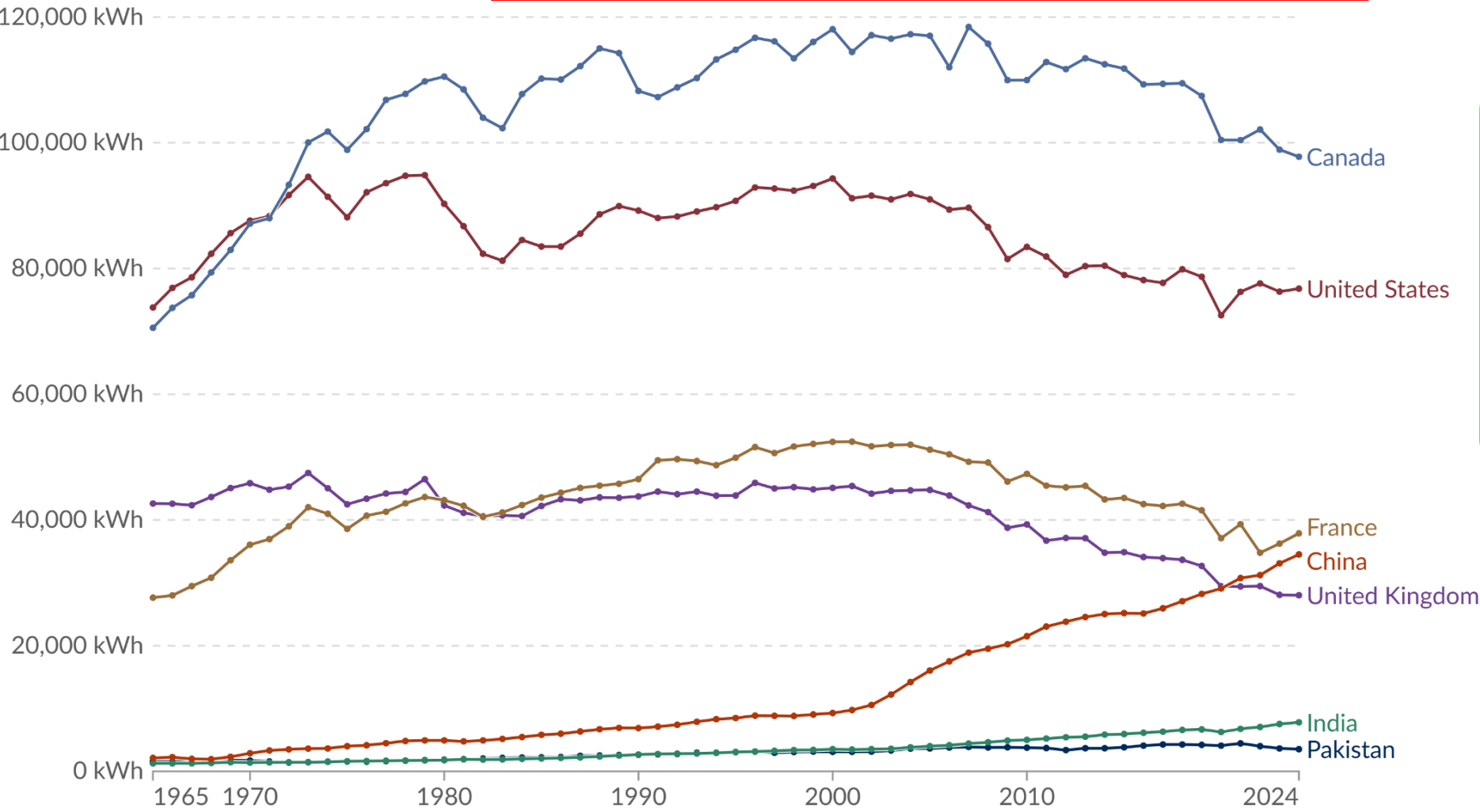
Data source: U.S. Energy Information Administration (2025) and other sources

OurWorldinData.org/energy | CC BY

Note: GDP data is expressed in international-\$⁴ at 2021 prices.

Energy use per person

Measured in kilowatt-hours¹ per person. Here, energy refers to primary energy² using the substitution method³.

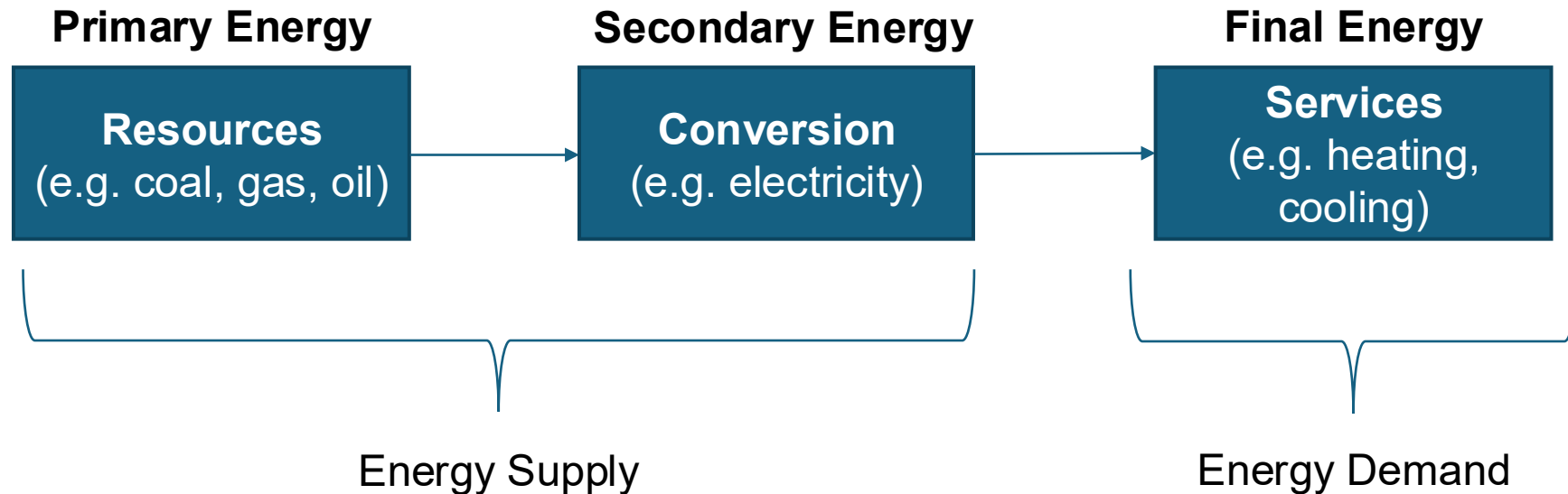


On average, by 2020, the average human has at its disposal, every year

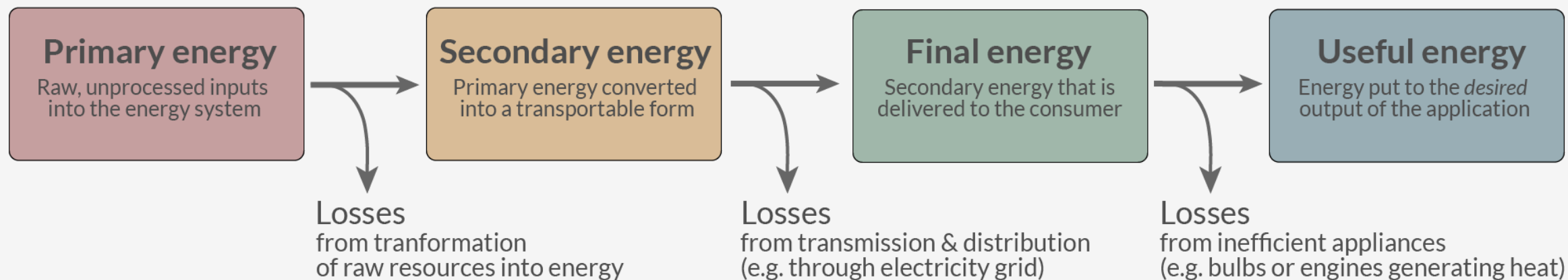
- 800 kg of crude oil, or
- 1.5 tons of coal, or
- 60 adults working non-stop day and night

Data source: U.S. Energy Information Administration (2025); Energy Institute - Statistical Review of World Energy (2025); Population based on various sources (2024)

Three main components of the energy system



The four ways of measuring energy



Example: Coal to power a lightbulb



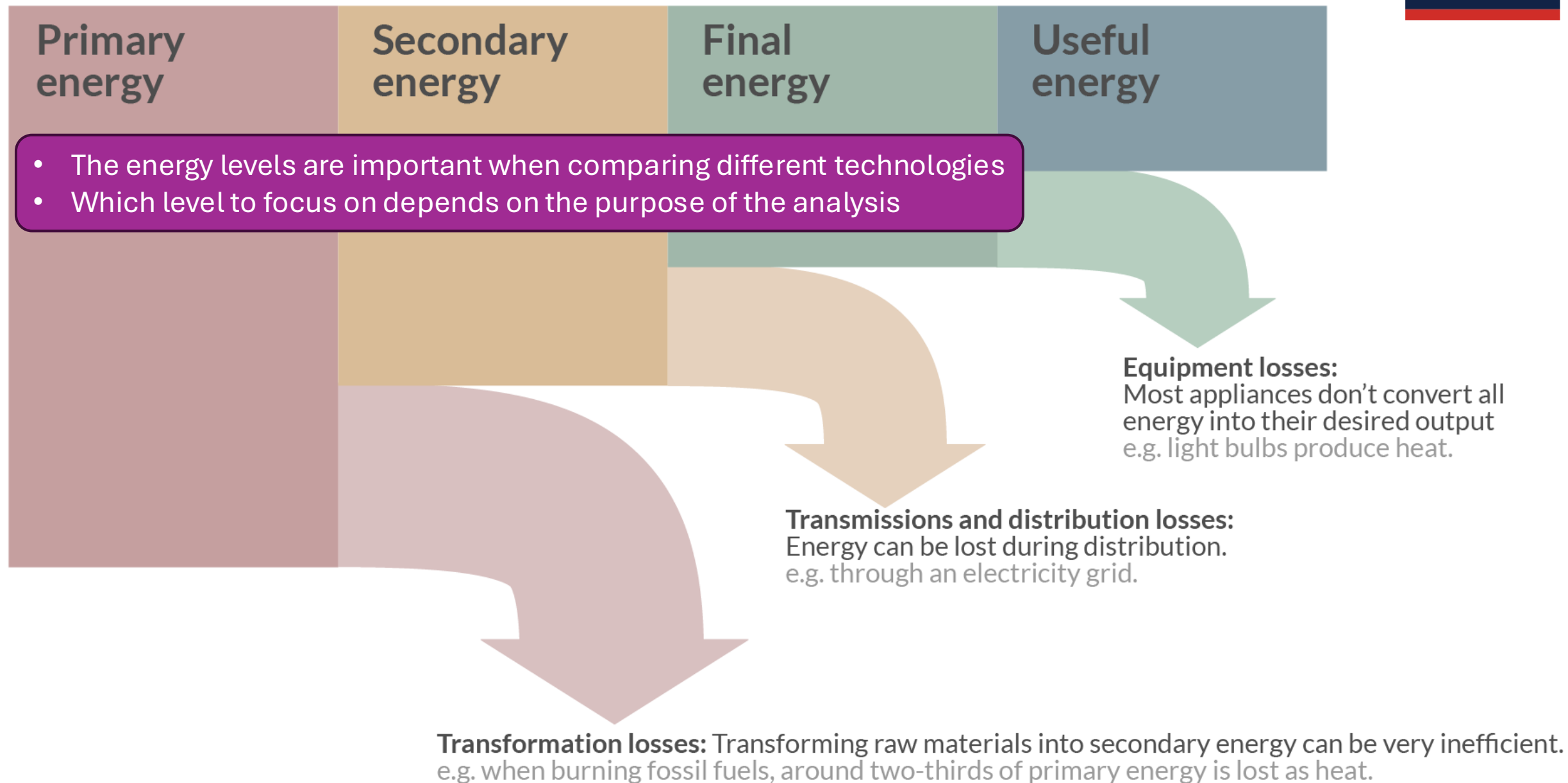
Example: Wood to provide heat

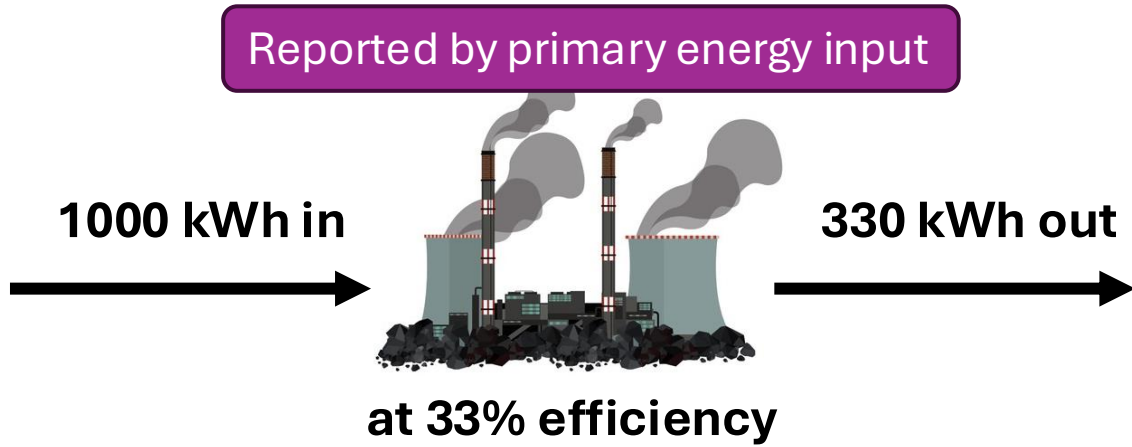


Example: Oil to drive a car



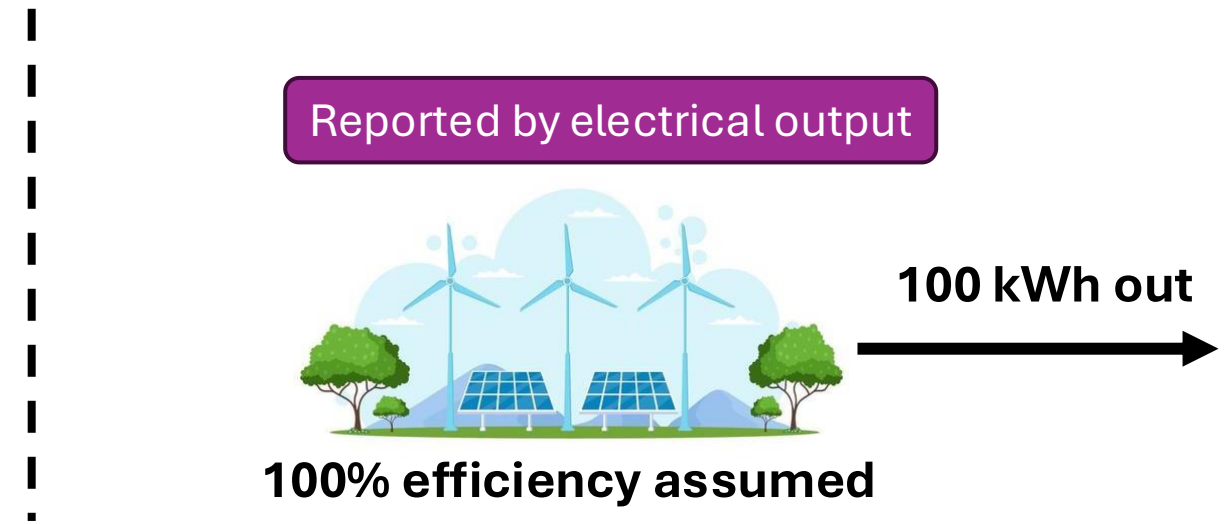
Useful energy is just a fraction of primary energy





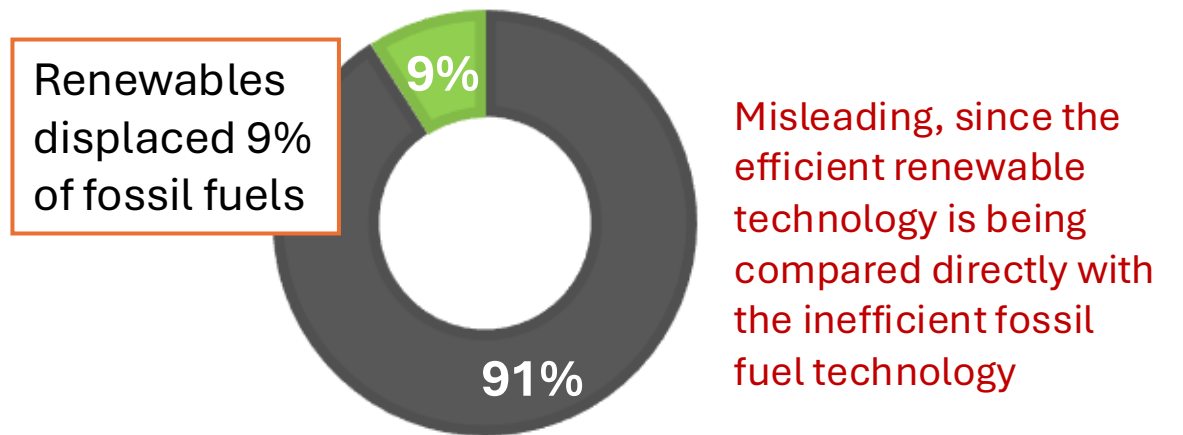
- Question: How much fossil fuels did the renewables replace?
- For this we must compare the primary energy inputs of both technologies

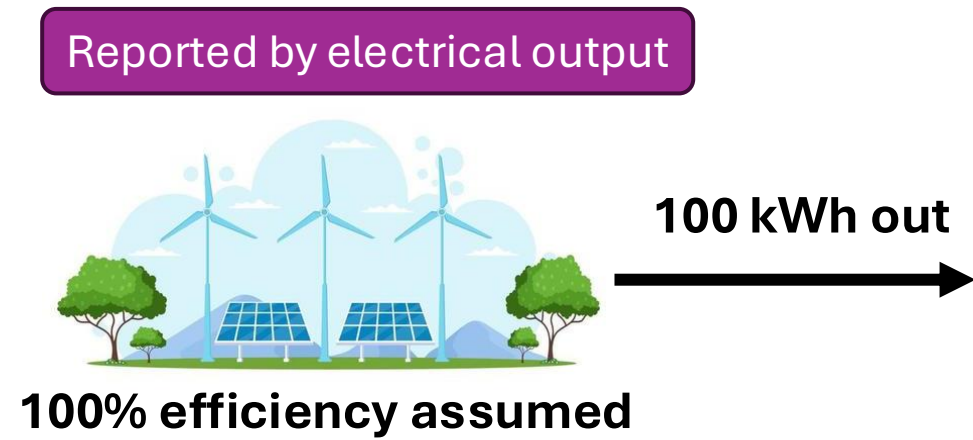
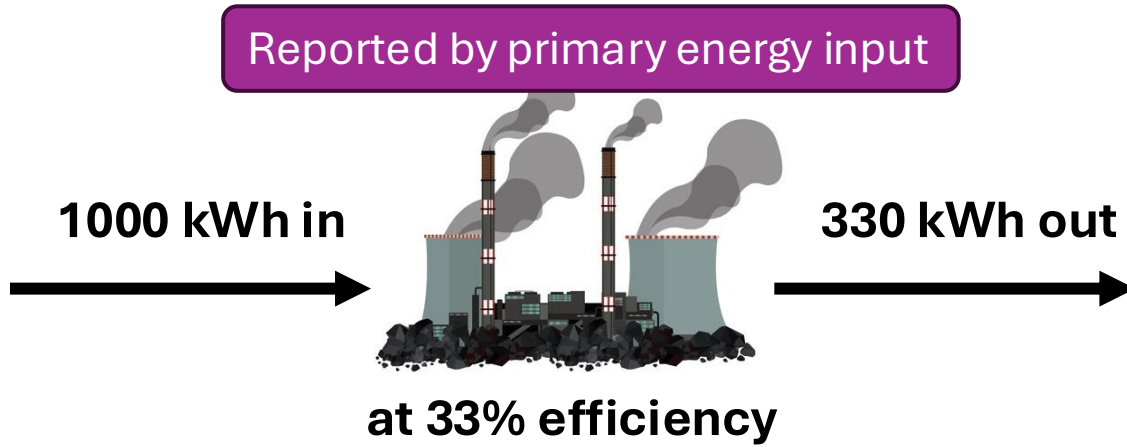
Source	Raw Energy Input (Primary Energy)
Coal and Gas (Fossil Fuels)	1000 kWh
Solar and Wind (Clean Power)	100 kWh
Total Primary Energy	1100 kWh



PRIMARY ENERGY SHARE (DIRECT METHOD)

■ Coal and Gas ■ Solar and Wind

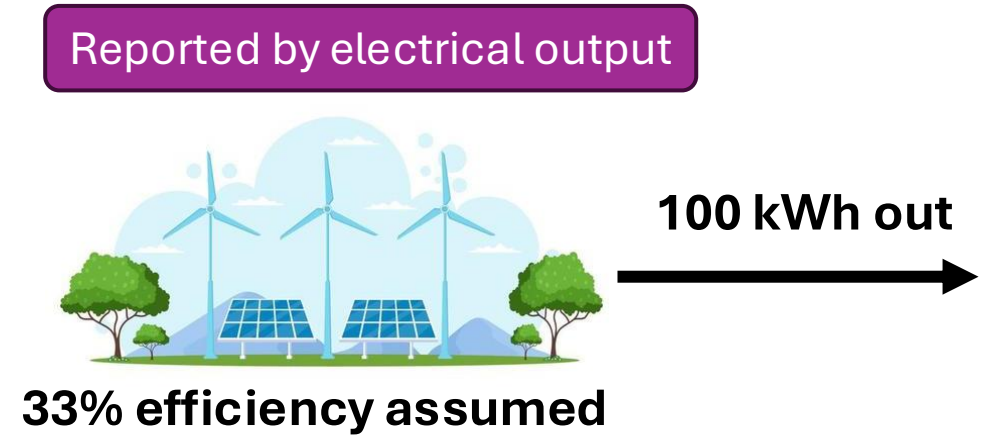
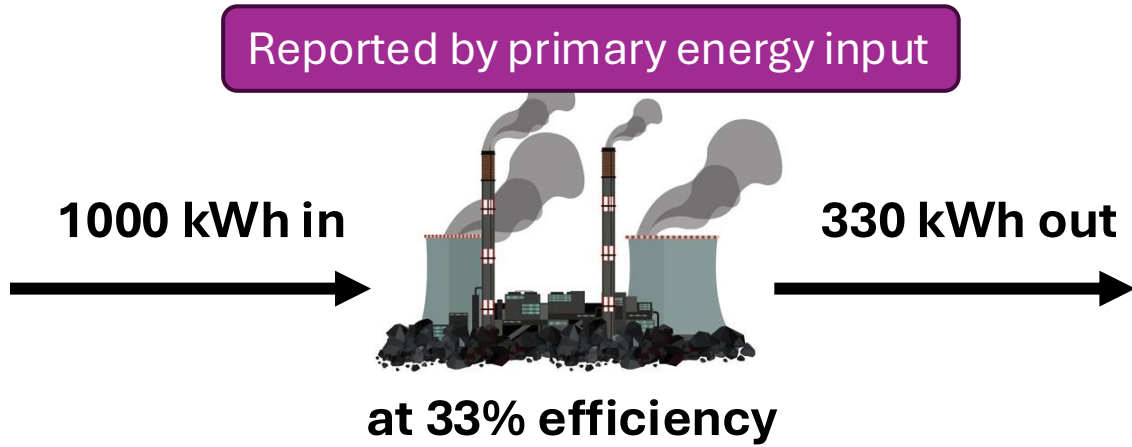




The Energy Substitution Method

Adjust non-fossil energy sources to the inputs that would be needed if it was generated from fossil fuels

$$\text{Primary Energy Input} = \frac{100 \text{ kWh}}{0.33} = 303 \text{ kWh}$$



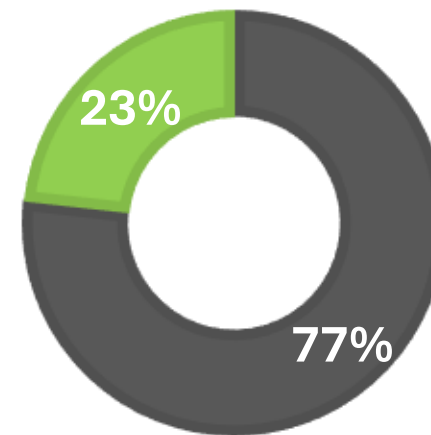
$$\text{Primary Energy Input} = \frac{100 \text{ kWh}}{0.33} = 303 \text{ kWh}$$

Efficiency Factor

PRIMARY ENERGY SHARE (SUBSTITUTION METHOD)

■ Coal and Gas ■ Solar and Wind

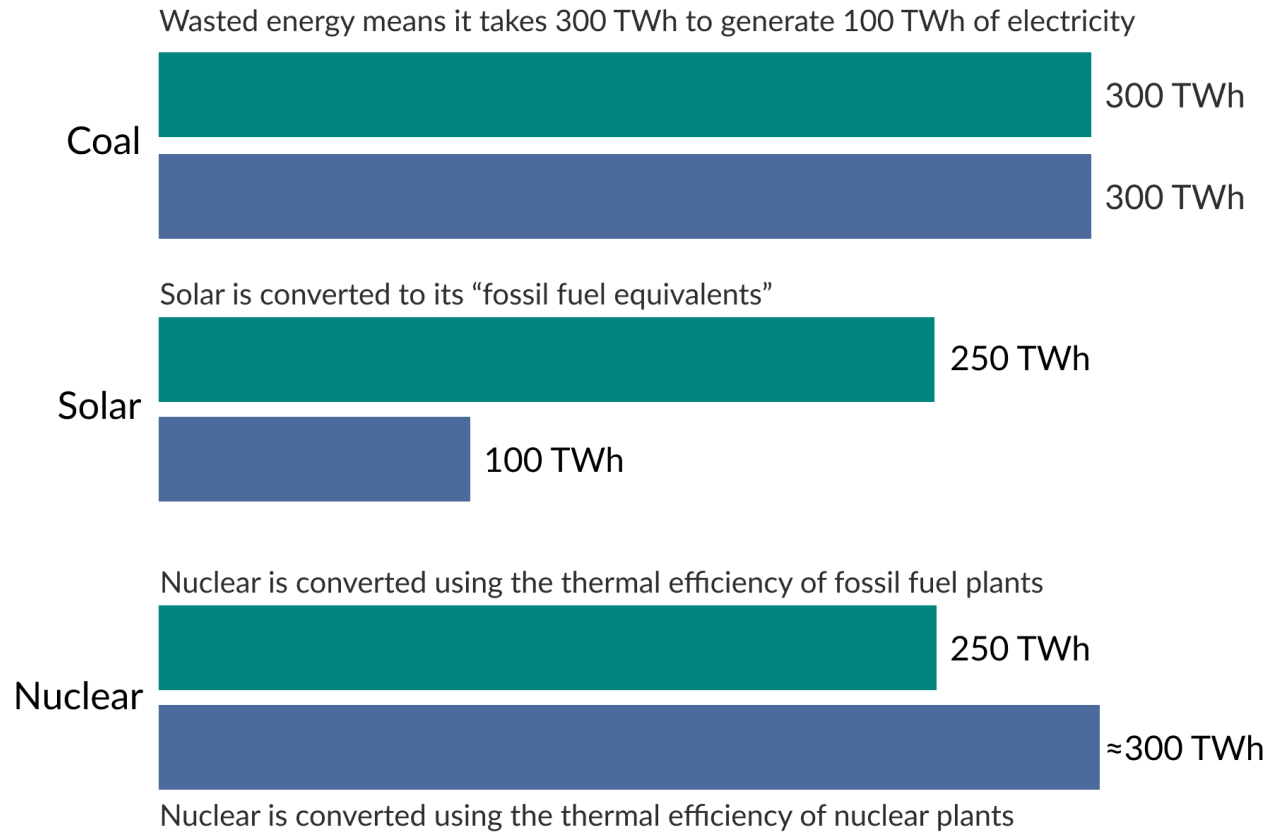
Source	Raw Energy Input (Primary Energy)
Coal and Gas (Fossil Fuels)	1000 kWh
Solar and Wind (Clean Power)	303 kWh
Total Primary Energy	1303 kWh



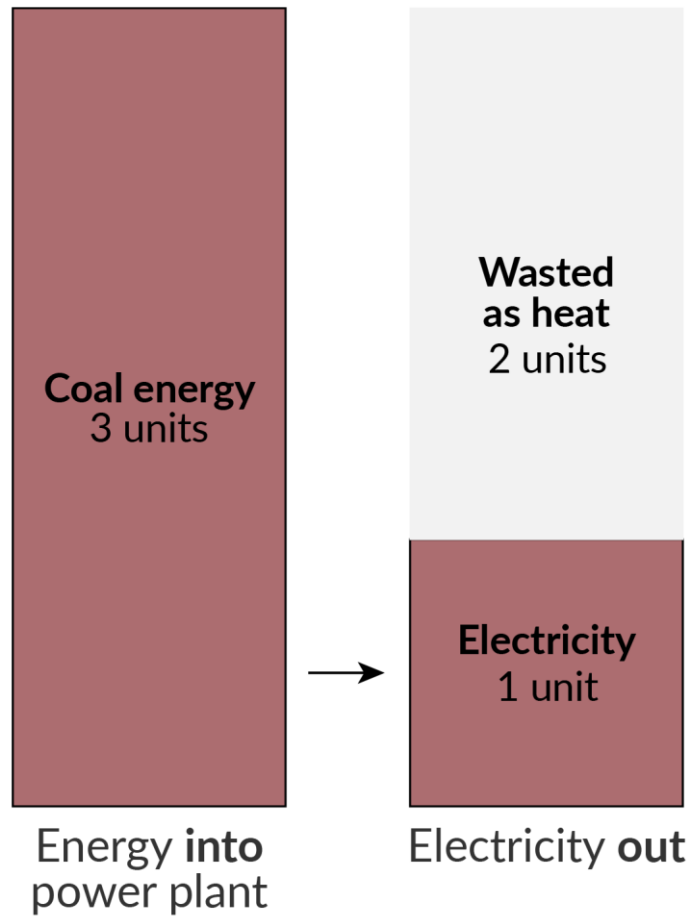
Renewables
displaced 23%
of fossil fuels

The new methodology: physical energy content

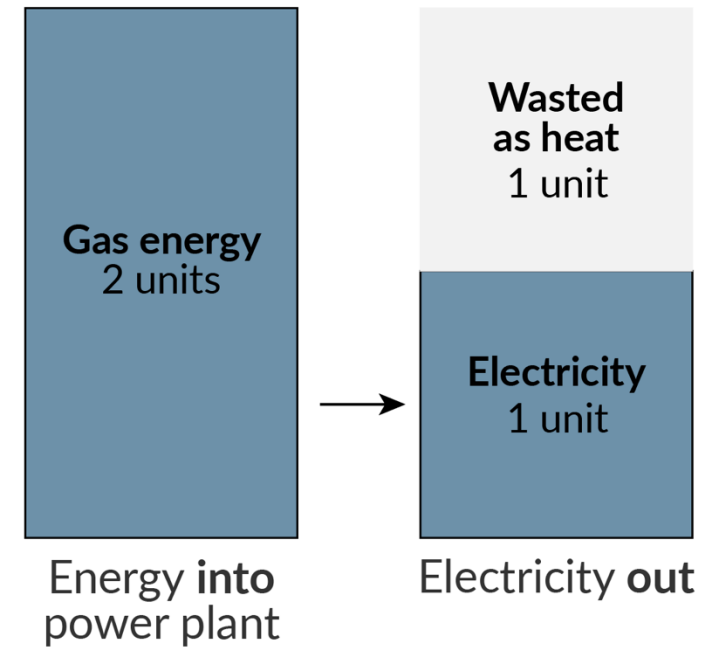
How 100 terawatt-hours (TWh) of electricity generation from different sources is counted in primary energy terms using the **substitution method** and **physical energy content method**



Coal plants
Average efficiency of 32%

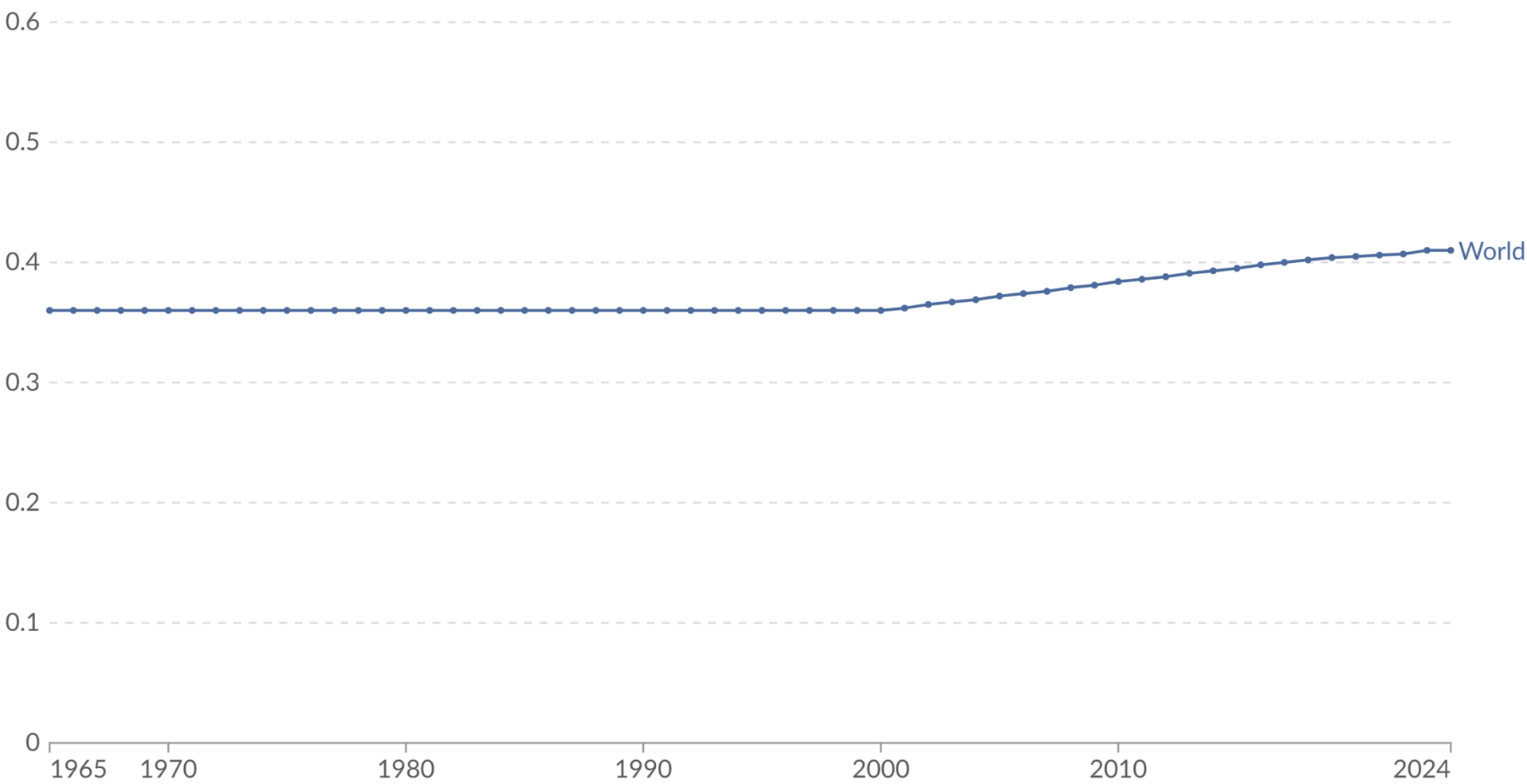


Gas plants
Average efficiency of 45%

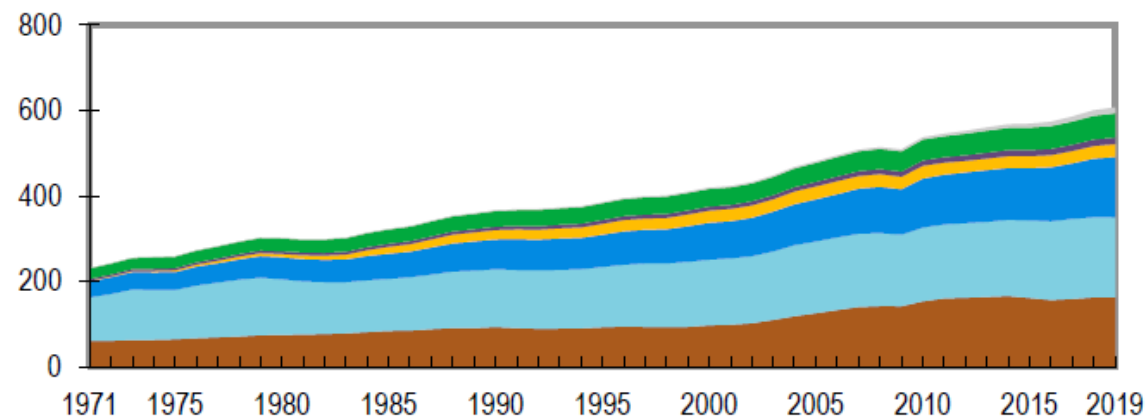


Thermal efficiency factor applied to non-fossil energy sources to convert them to primary energy equivalents, 1965 to 2024

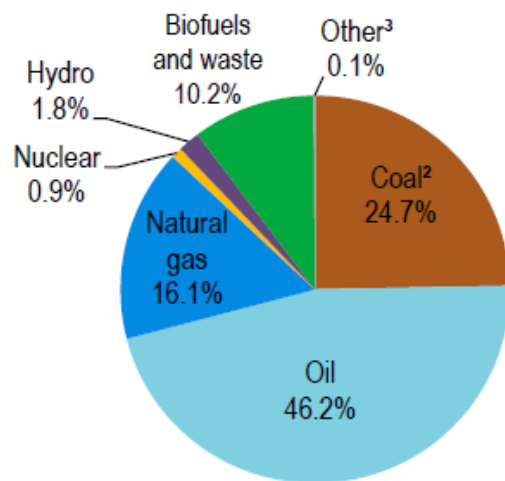
This is the factor that the Energy Institute applies to renewables and nuclear electricity outputs to convert them into 'primary energy equivalents' to compare them to energy supply from fossil fuels. For example, solar electricity output would be divided by the efficiency factor in a given year to convert it into 'primary energy equivalents'.



World¹ total energy supply by source, 1971-2019 (EJ)

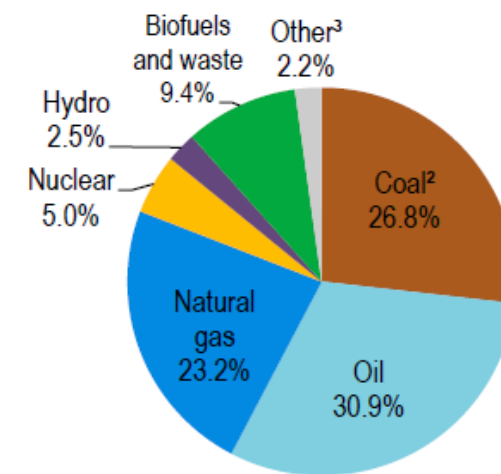


1973



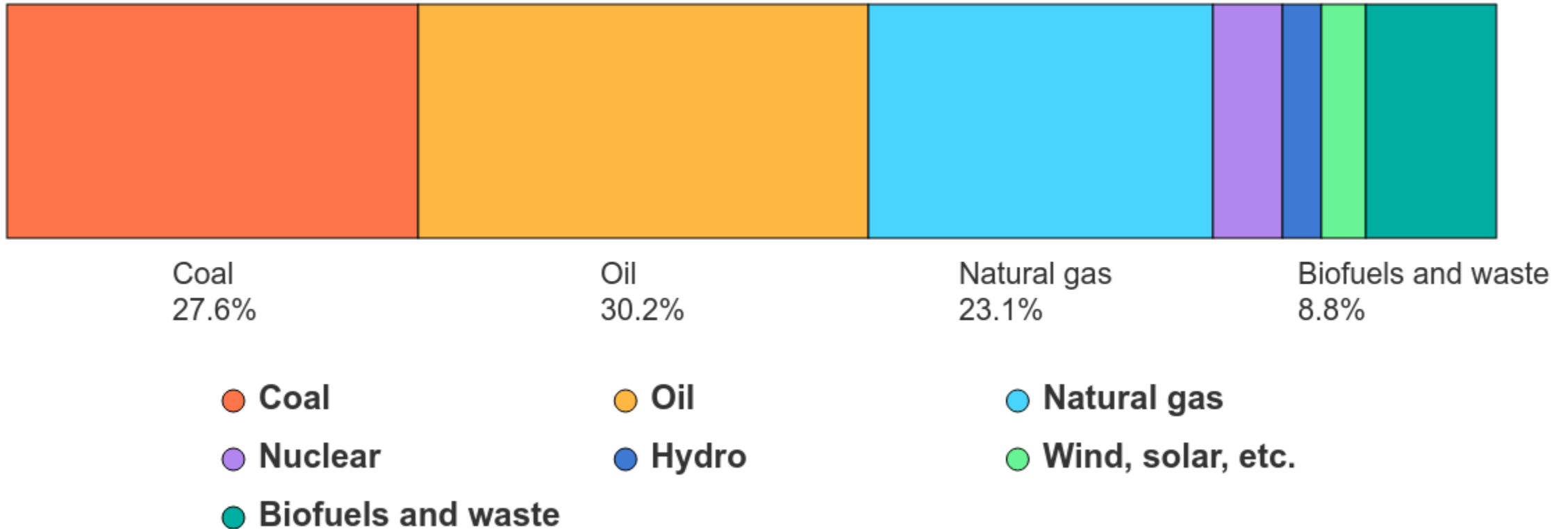
254 EJ

2019



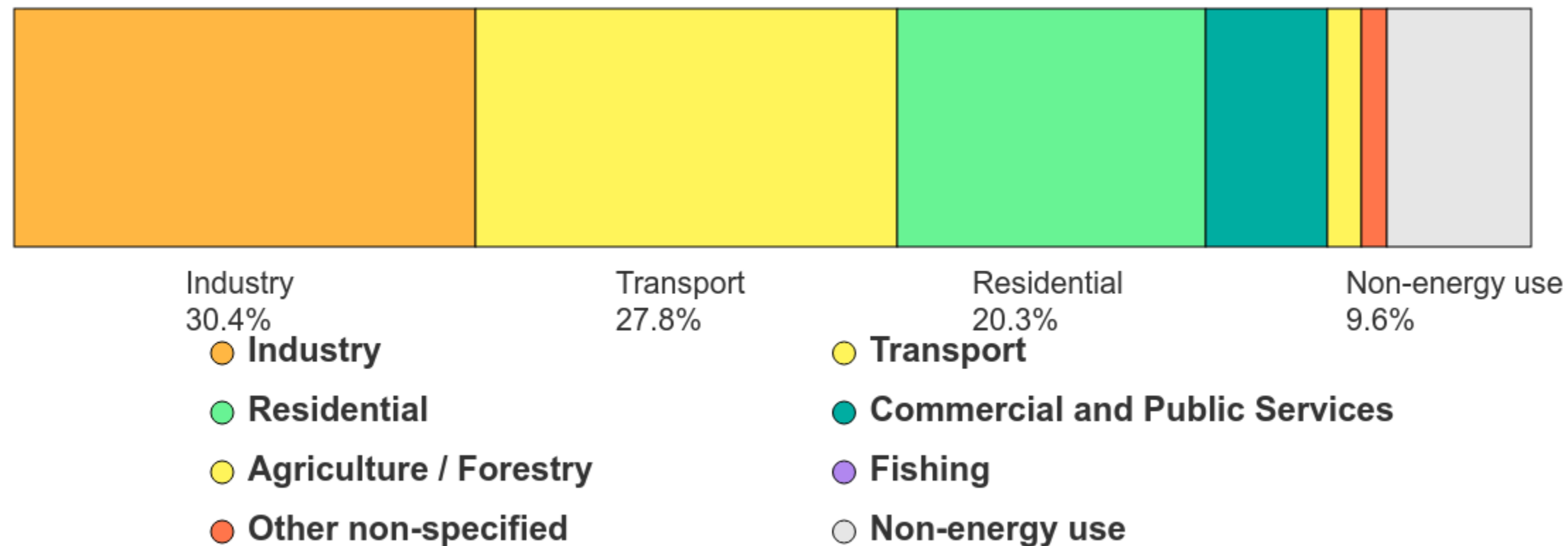
606 EJ

Total energy supply by source, World, 2022



Source: International Energy Agency. Licence: CC BY 4.0

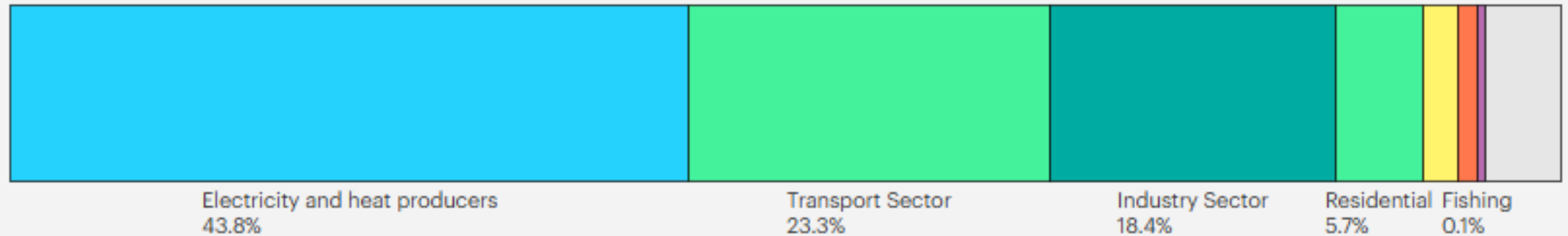
Total final consumption, World, 2022



Source: International Energy Agency. Licence: CC BY 4.0

CO2 emissions by sector, World, 2022

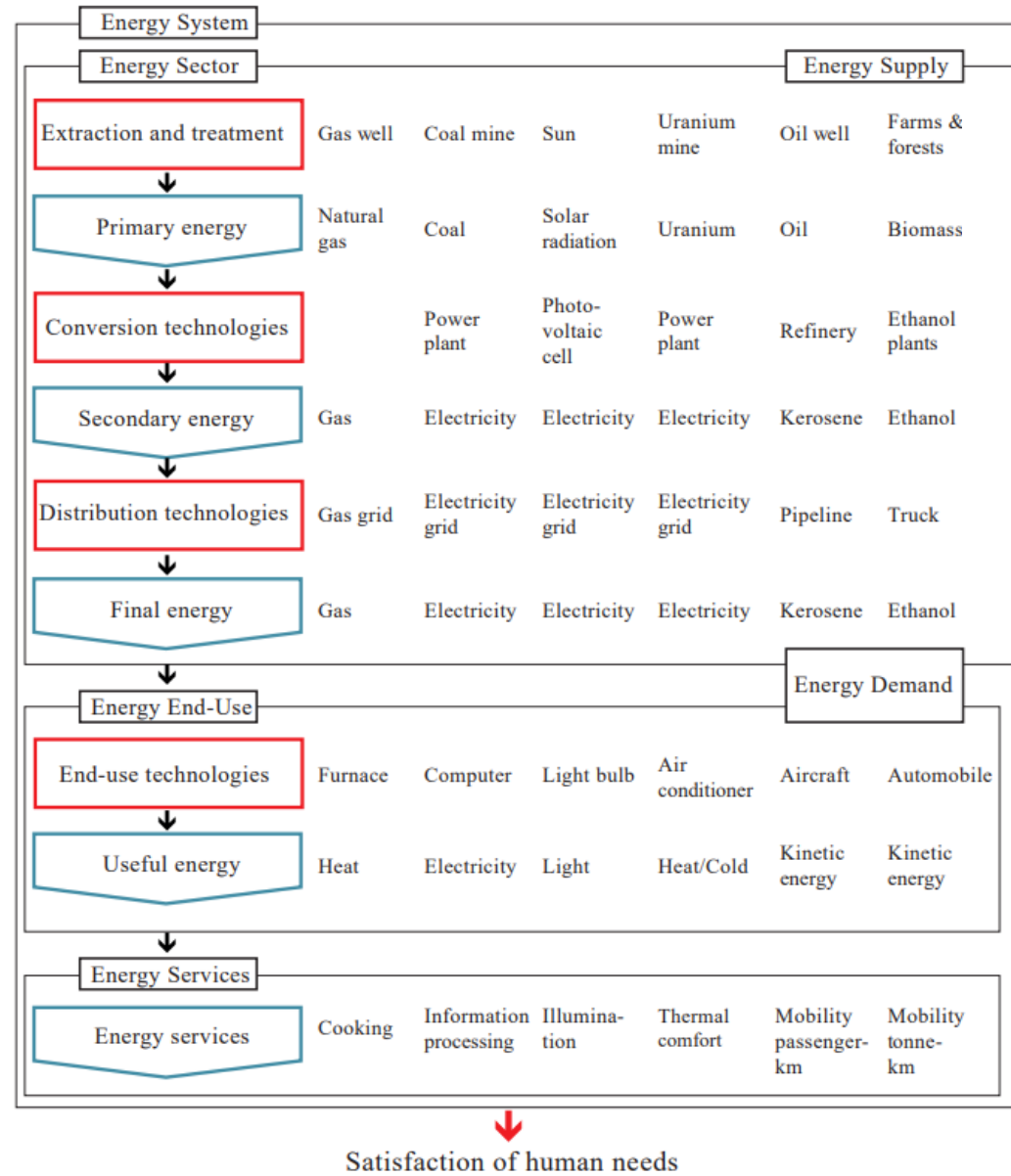
CO2 emissions by sector, World, 2022



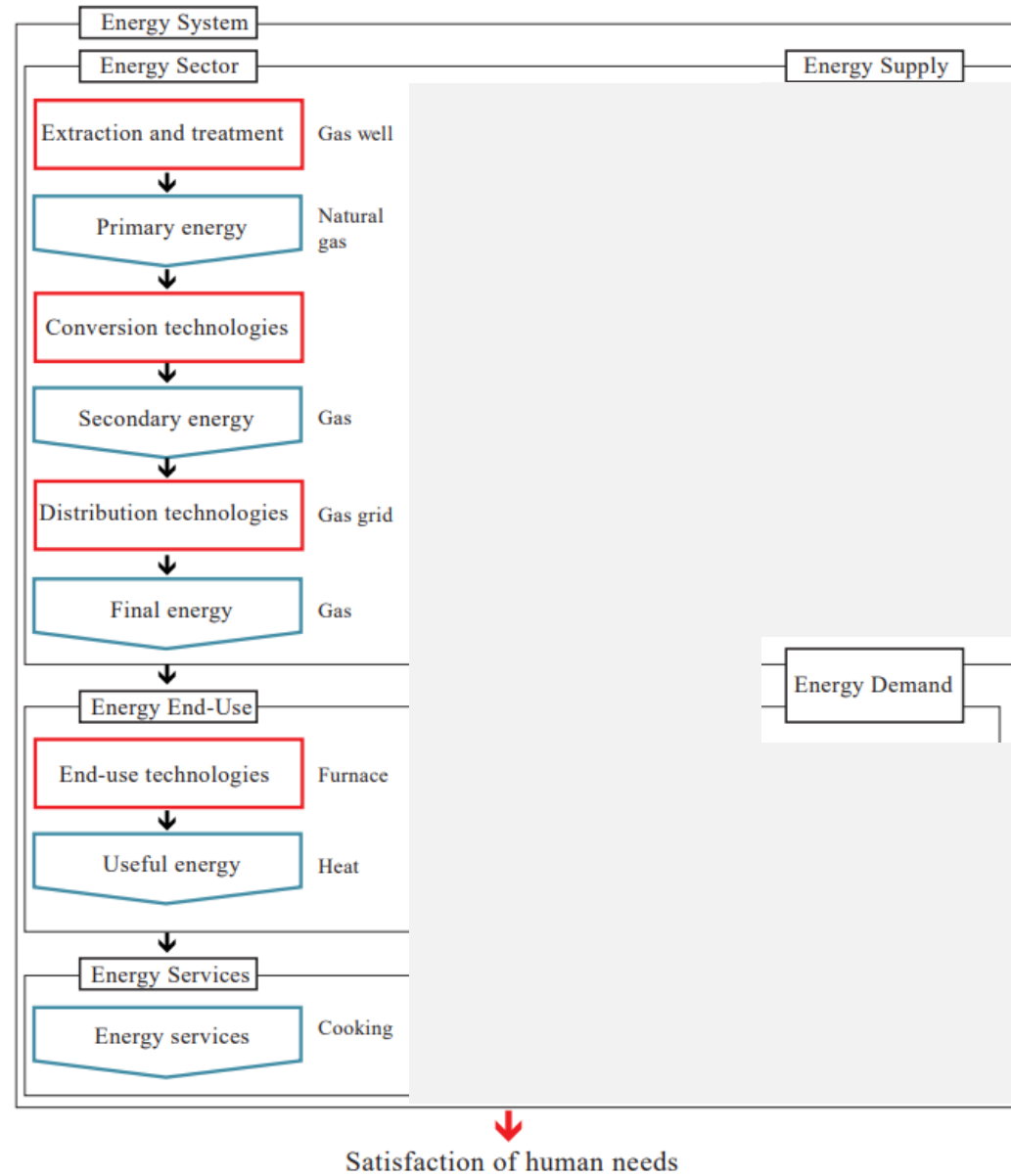
<https://www.iea.org/world/emissions>

Source: International Energy Agency. Licence: CC BY 4.0

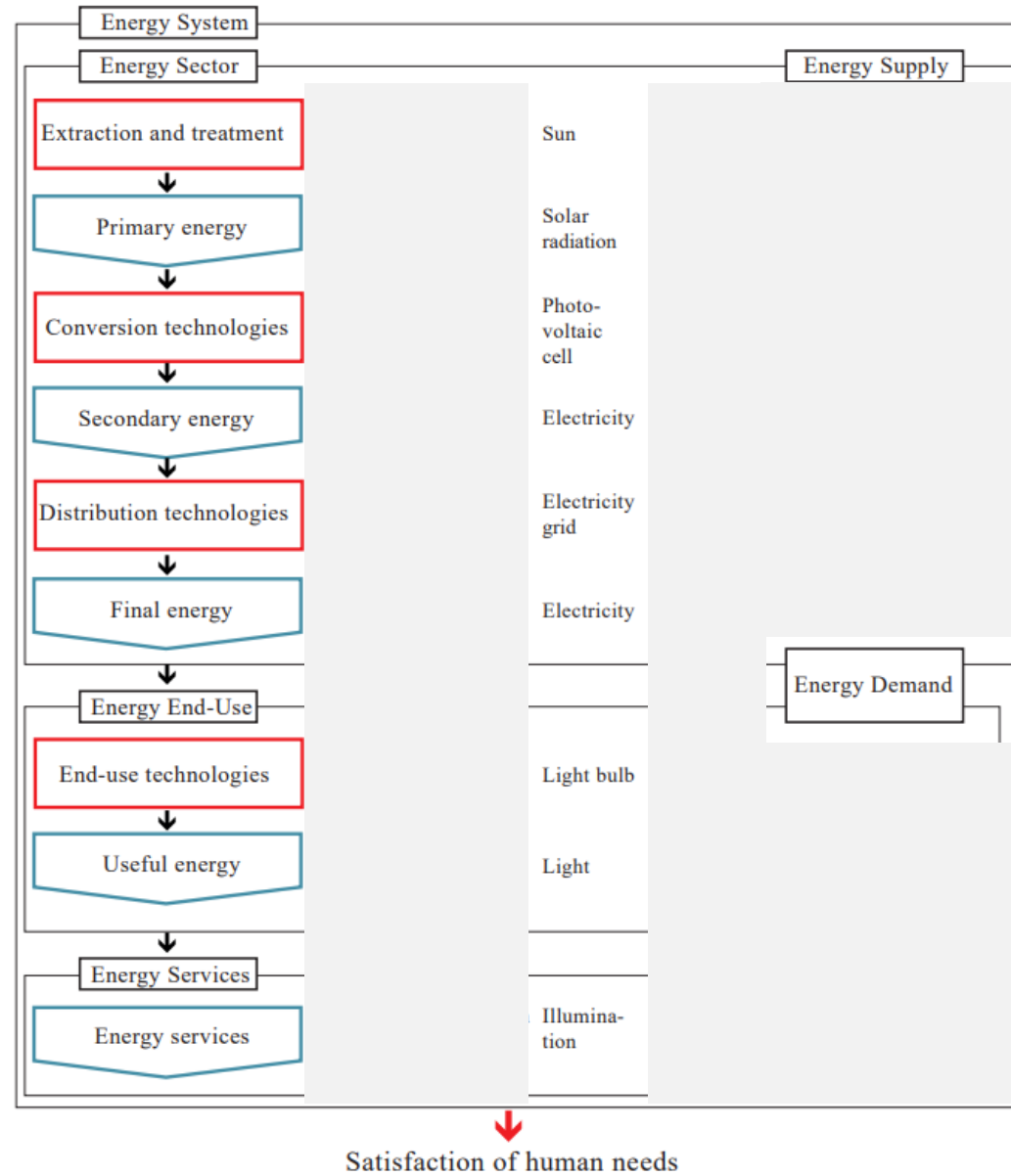
Components of and processes in the energy system: Some details



Components of and processes in the energy system: Example starting with a fossil resource

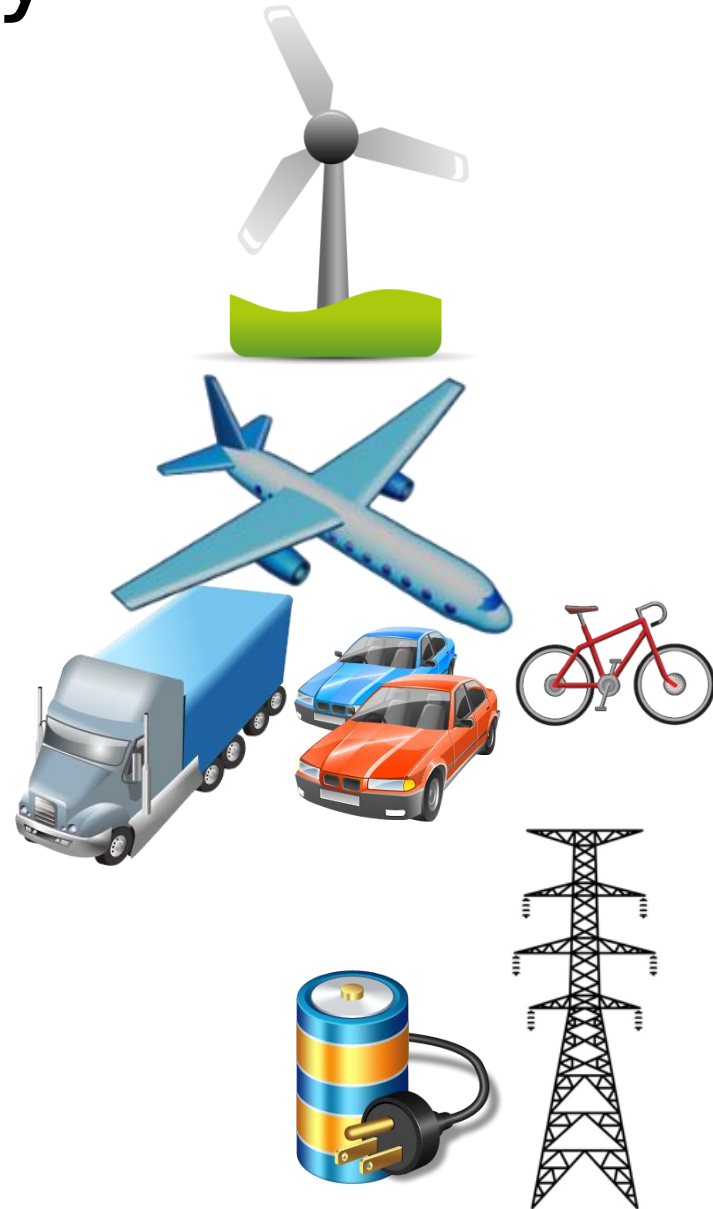


Components of and processes in the energy system: Example starting with a renewable resource

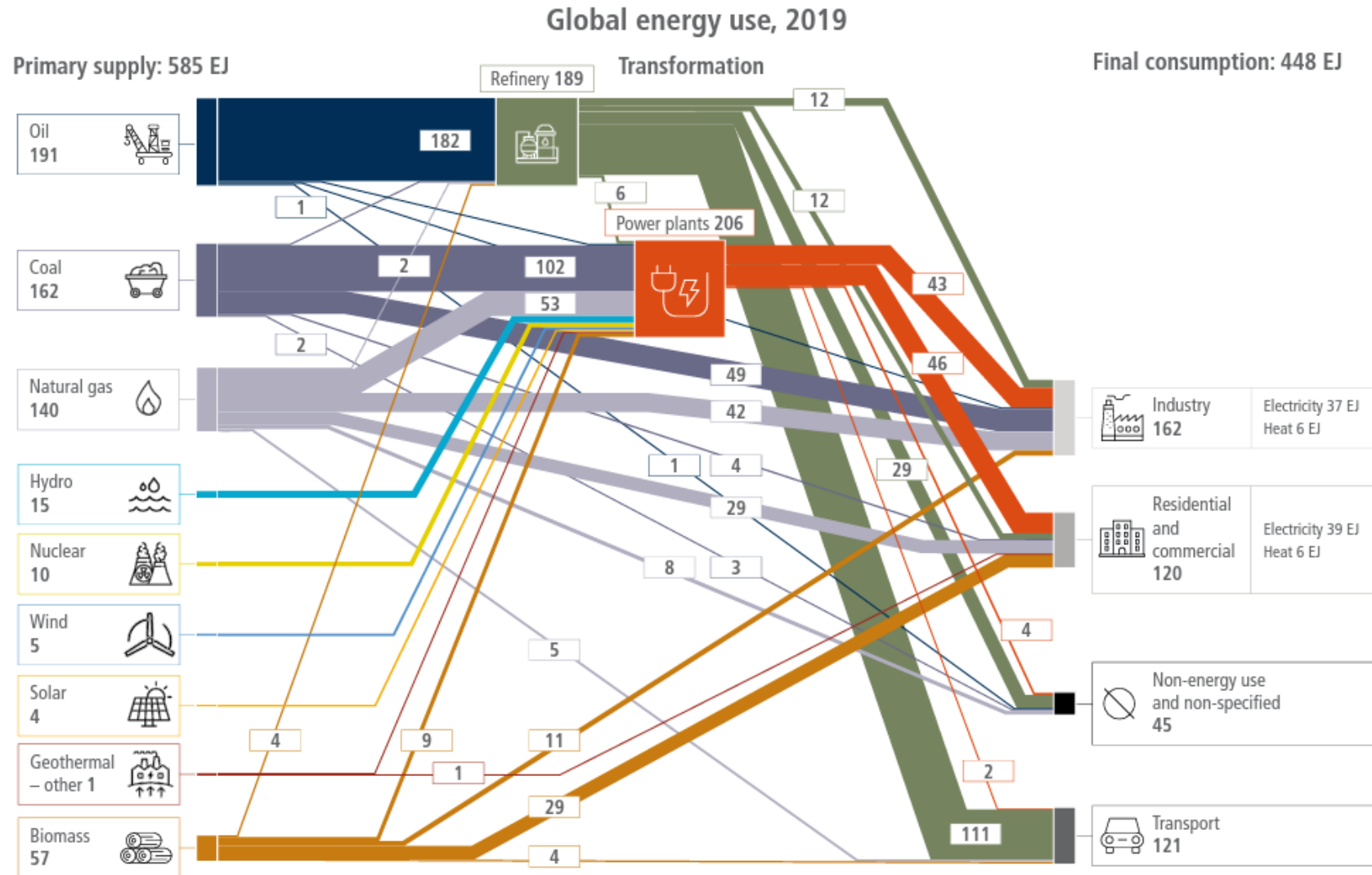


A Generic Mitigation Strategy

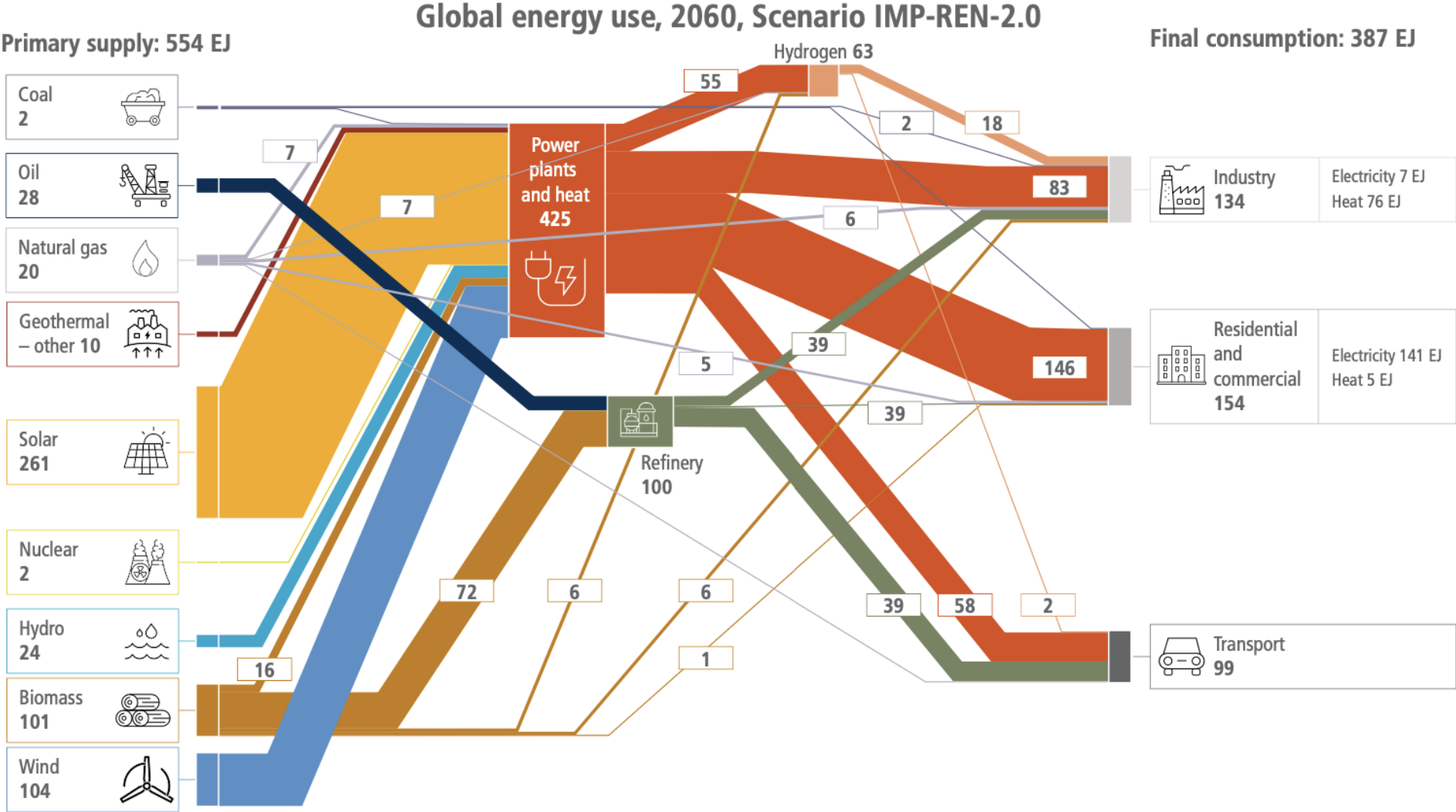
1. Decarbonize Energy Supply
2. Reduce Energy Demand
3. Switch to Low-Carbon Fuels



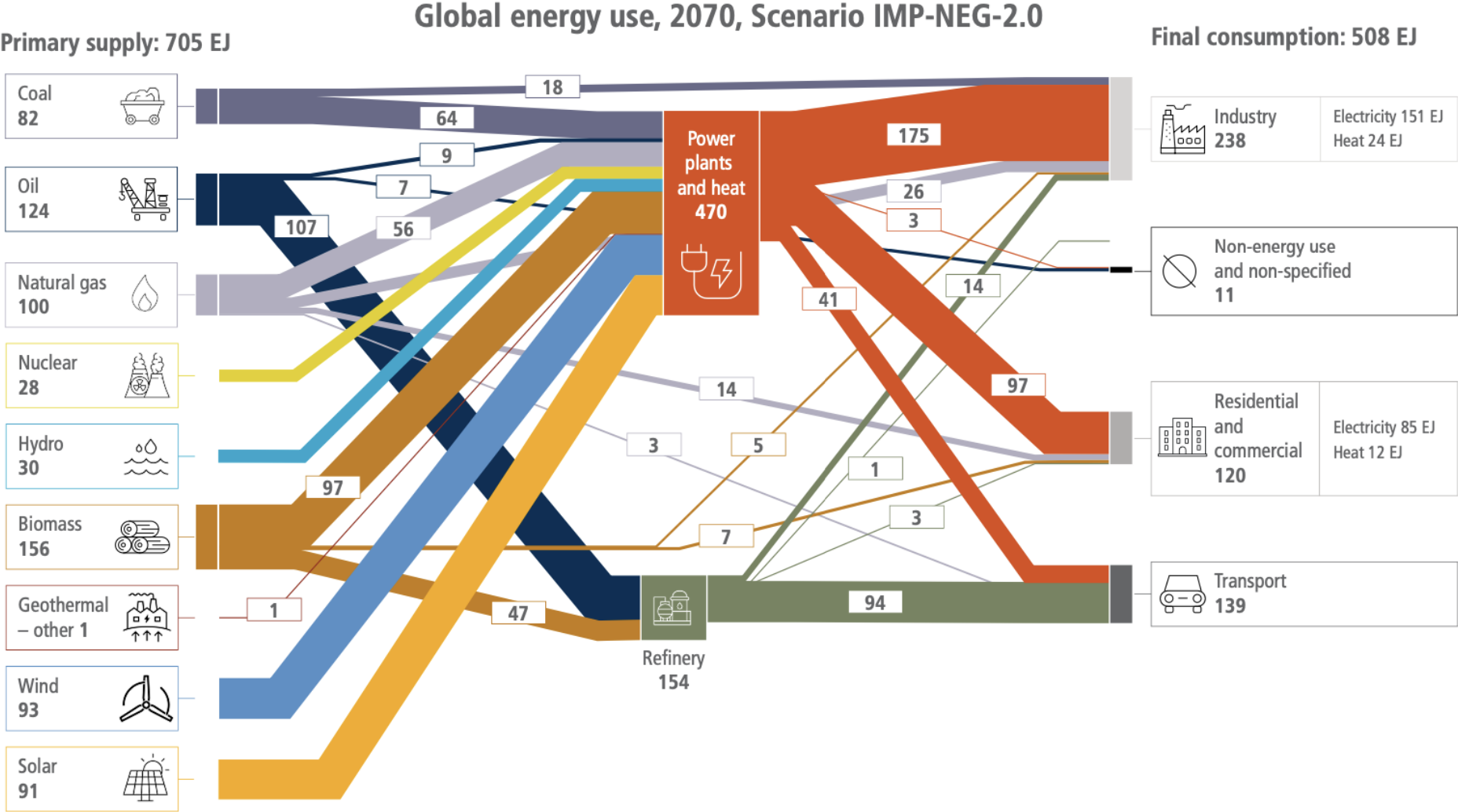
Global flow of energy in 2019



If the world successfully limits warming to well below 20C, the energy systems of the future will be different



If the world successfully limits warming to well below 20C, the energy systems of the future will be different



Quick refresher #1: Energy Units

Energy Numbers in Context

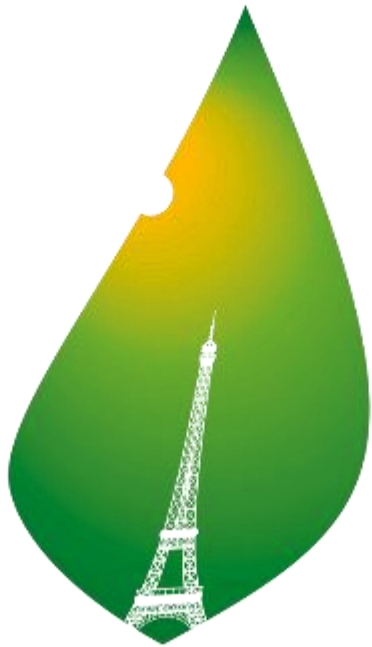
Energy Quantity	Handy Equivalent	What It Represents
1 PJ (10^{15} J)	≈ 278 GWh	Electricity for $\approx 30,000$ homes/year
	≈ 24 million L petrol ($\sim 150,000$ barrels)	Fuel for $\approx 40,000$ cars (15,000 km each)
	≈ 10 days of a 500 MW plant	—
	≈ 24 ktoe (thousand tonnes oil equivalent)	—
1 EJ (10^{18} J)	≈ 277 TWh	$\approx 1.8/2$ times of pakistan's total electricity use
	≈ 25 Mt coal or 17 bcm natural gas	Equivalent fuels releasing ~ 1 EJ
	≈ 170 million barrels oil	≈ 30 M U.S. homes powered/year
	≈ 24 Mtoe (million tonnes oil equivalent)	—

Quick pointers

- 1 PJ = city or industrial-scale energy use.
- 1 EJ = national-scale energy use.
- World total (2023): ≈ 620 EJ/yr.
- Pakistan (2023): ≈ 3.9 EJ/yr.

Quick refresher #2: International climate policy

The international community has established ambitious long-term goals



COP21 • CMP11
PARIS 2015
UN CLIMATE CHANGE CONFERENCE

*Holding the increase in the global average temperature to **well below 2 °C** above pre-industrial levels and to pursue efforts to limit the temperature increase to **1.5 °C** above pre-industrial levels*

How much do we need
to reduce global GHG emissions
to implement Paris?

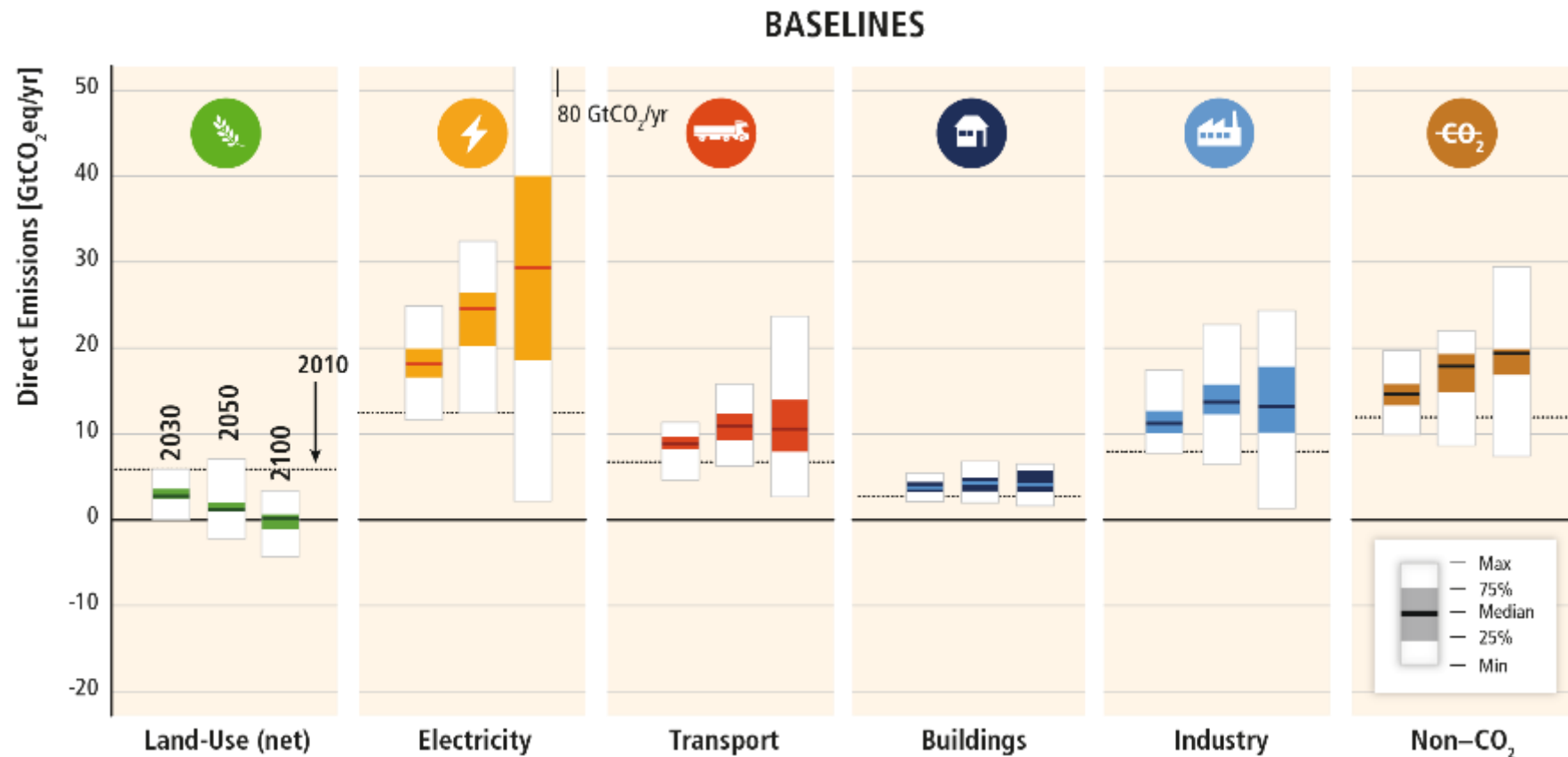
100%

How much should energy-system emissions be reduced?

100% or
more

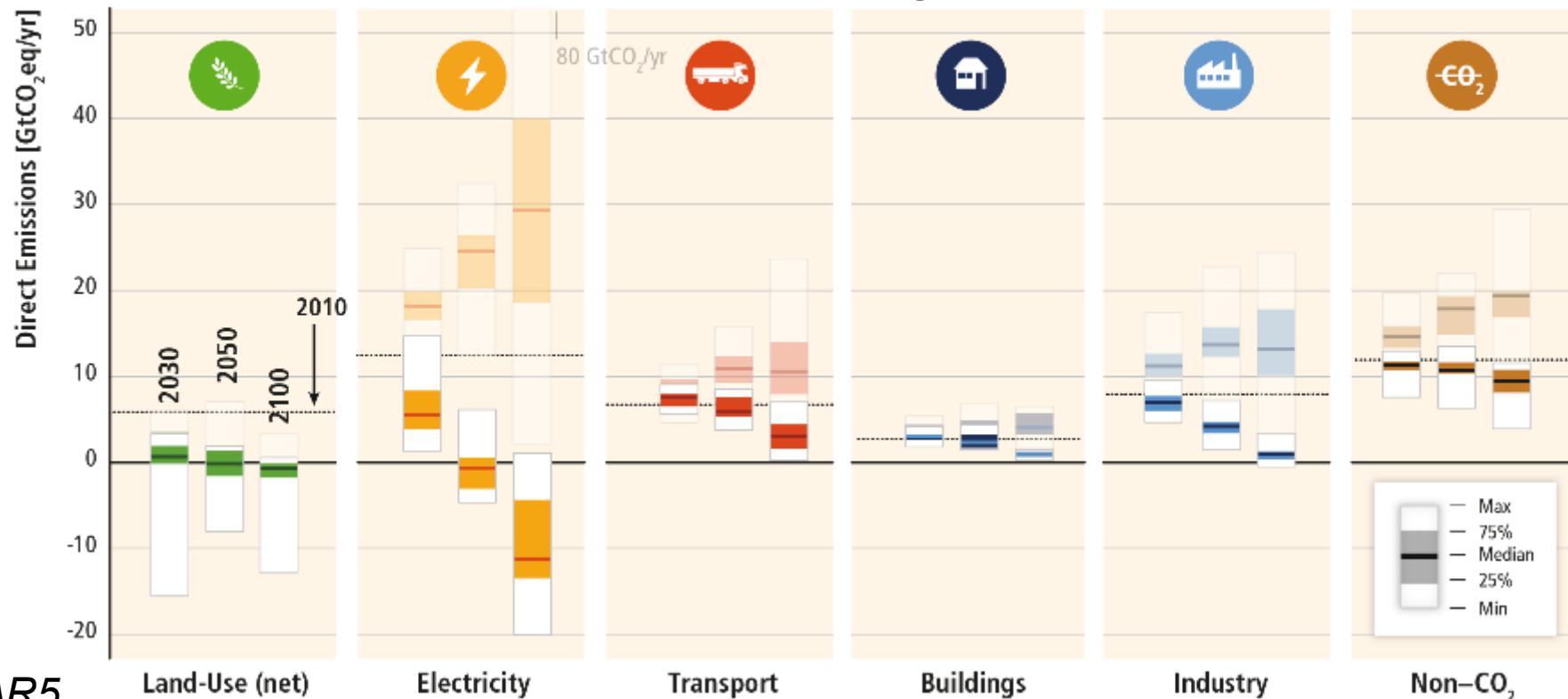
Scenarios produced by IAMs suggest emission increases in the absence of focused policies

GHG emissions in “No climate Policy” scenarios



Scenarios based on IAMs that incorporate Paris goals are characterized by major emissions reductions in the energy system

GHG emissions in mitigation scenarios (dark) compared to emissions in baseline scenarios (light)



Source: IPCC, AR5

Quick refresher #3: Energy technologies

What kinds of energy technologies result in non-zero emissions?

What kinds of energy technologies result in non-zero emissions?

Coal-fired power plants



Oil refineries, industries



Combustion engines



Building furnaces



What kinds of energy technologies result in low- or zero- emissions?

What kinds of energy technologies result in low- or zero- emissions?

Renewables



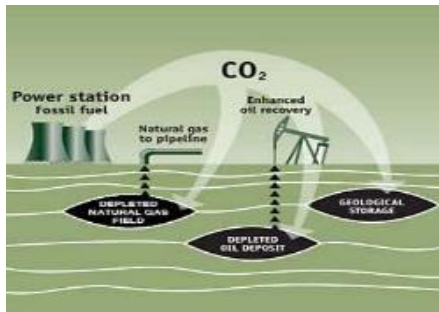
Nuclear



Biofuels



**Carbon Capture
Utilization &
Storage**



Hydrogen



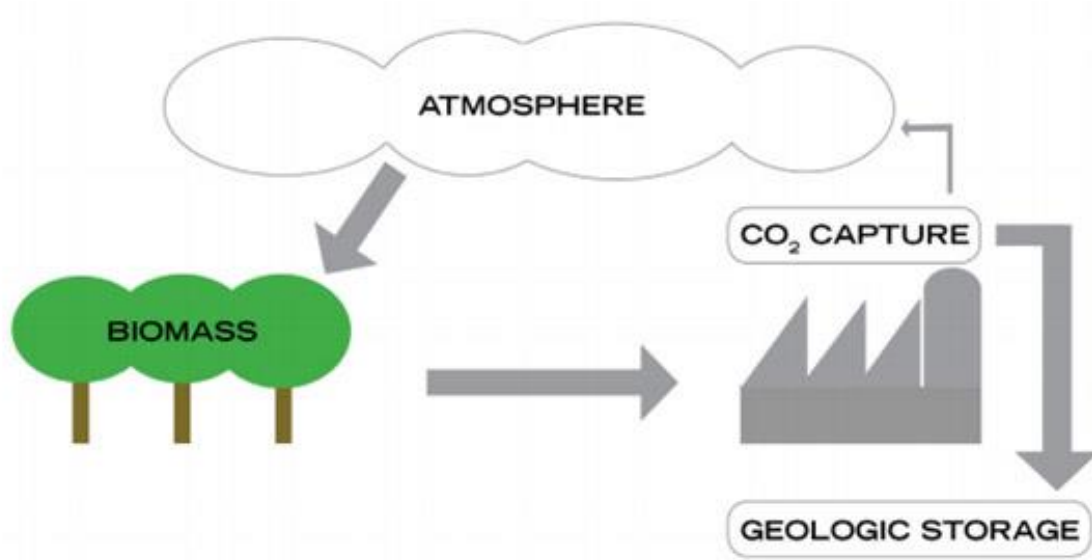
Electric vehicles



What kinds of energy technologies result in negative emissions?

What kinds of energy technologies result in negative emissions?

Bioenergy + CCUS



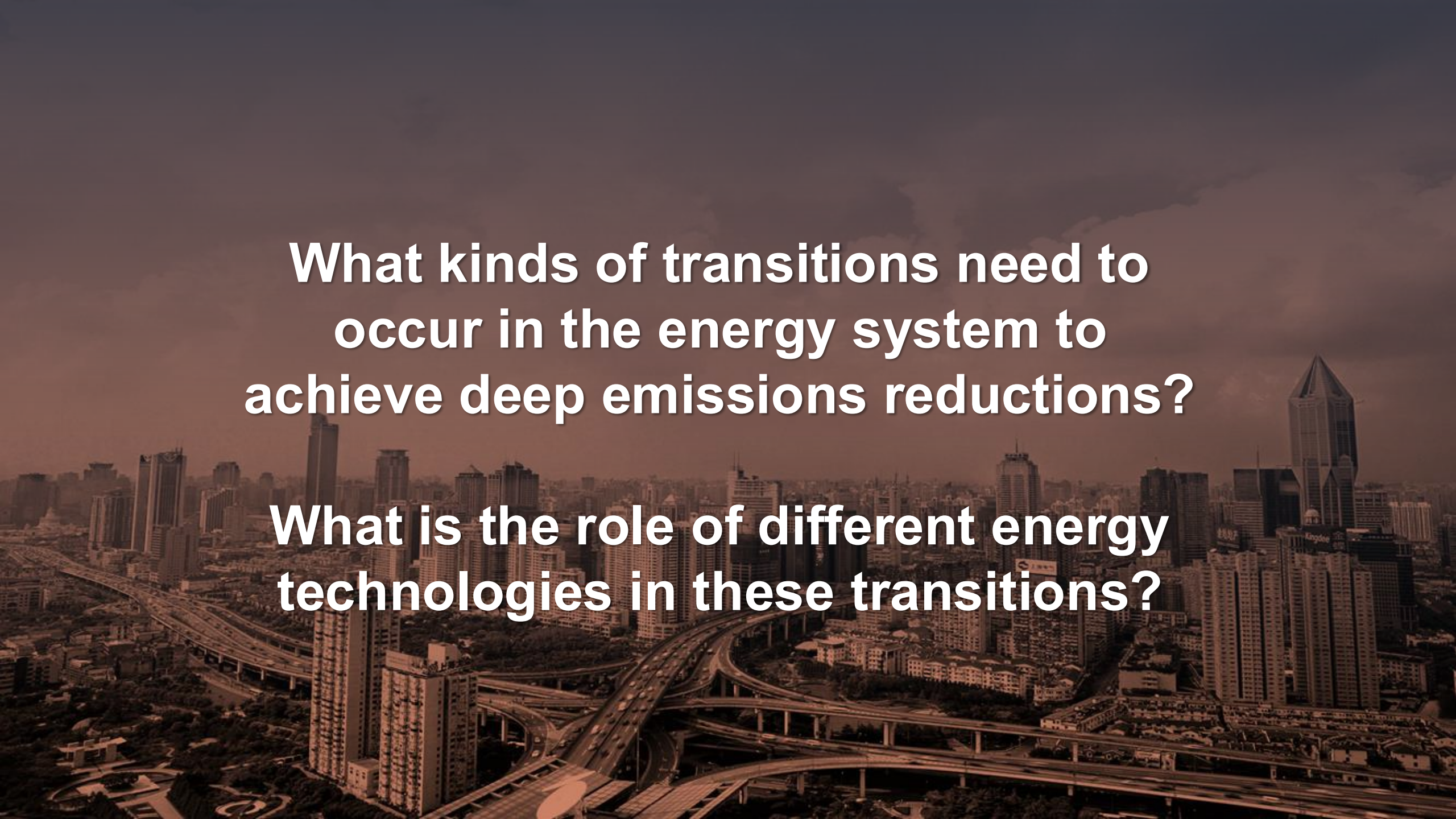
Direct Air Capture



<https://www.globalccsinstitute.com/insights/authors/AliceGibson/2015/11/25/importance-bio-ccs-deliver-negative-emissions>

Key questions of interest in the energy-climate problem that we will focus on today

- What kinds of transitions need to occur in the energy system to achieve deep emissions reductions? What is the role of different energy technologies in these transitions?
- What is the role of institutions in those transitions?

An aerial photograph of a city, likely Hong Kong, showing a dense urban landscape with numerous skyscrapers and a complex network of elevated highways and bridges in the foreground. The image has a warm, sepia-toned filter. Two text questions are overlaid on the image in a bold, white, sans-serif font.

**What kinds of transitions need to
occur in the energy system to
achieve deep emissions reductions?**

**What is the role of different energy
technologies in these transitions?**

Scenarios consistent with Paris goals are characterized by major scaling up of low carbon technologies

Boxes indicate 25th and 75th percentiles;

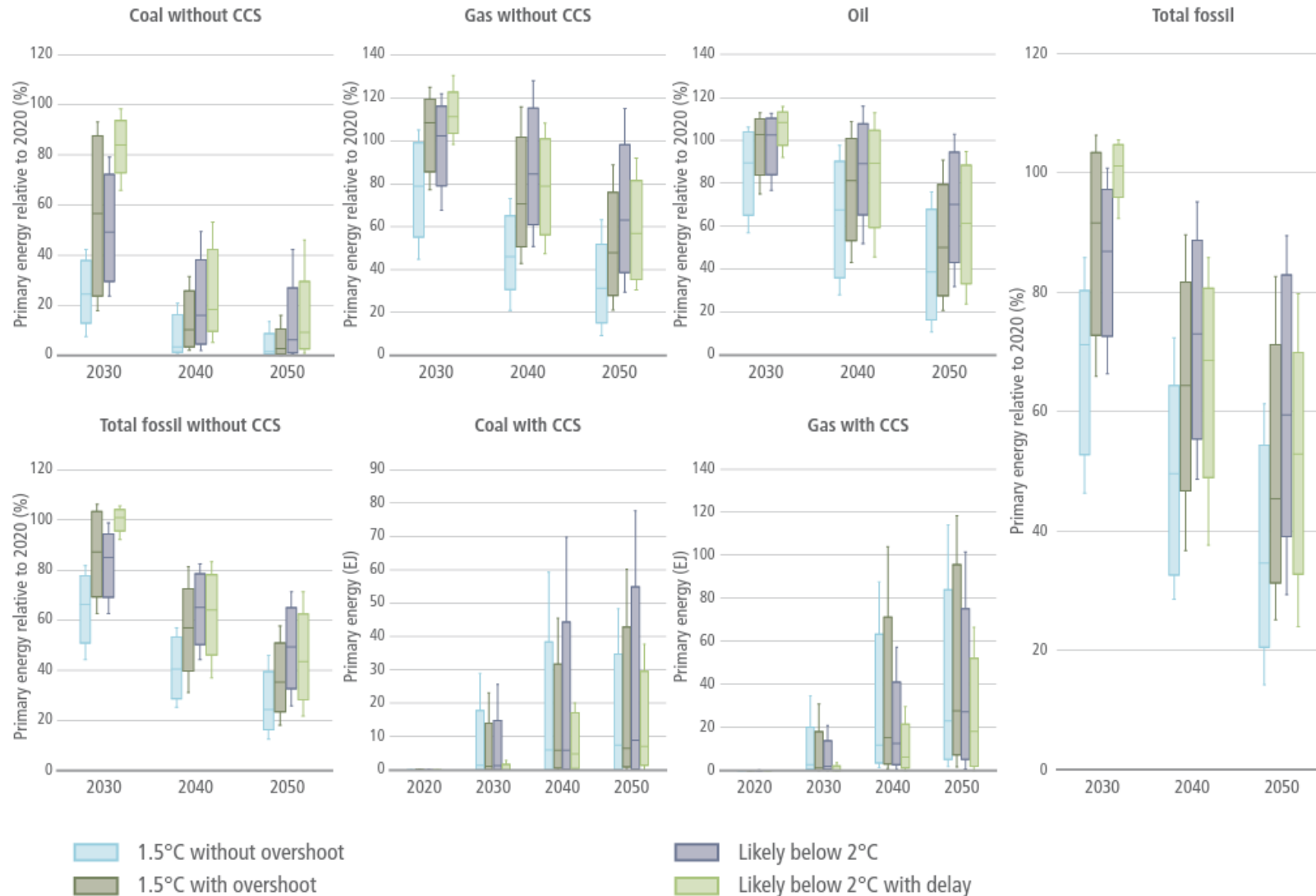
whiskers indicate 5th and 95th percentiles

Source:
IPCC,
AR6

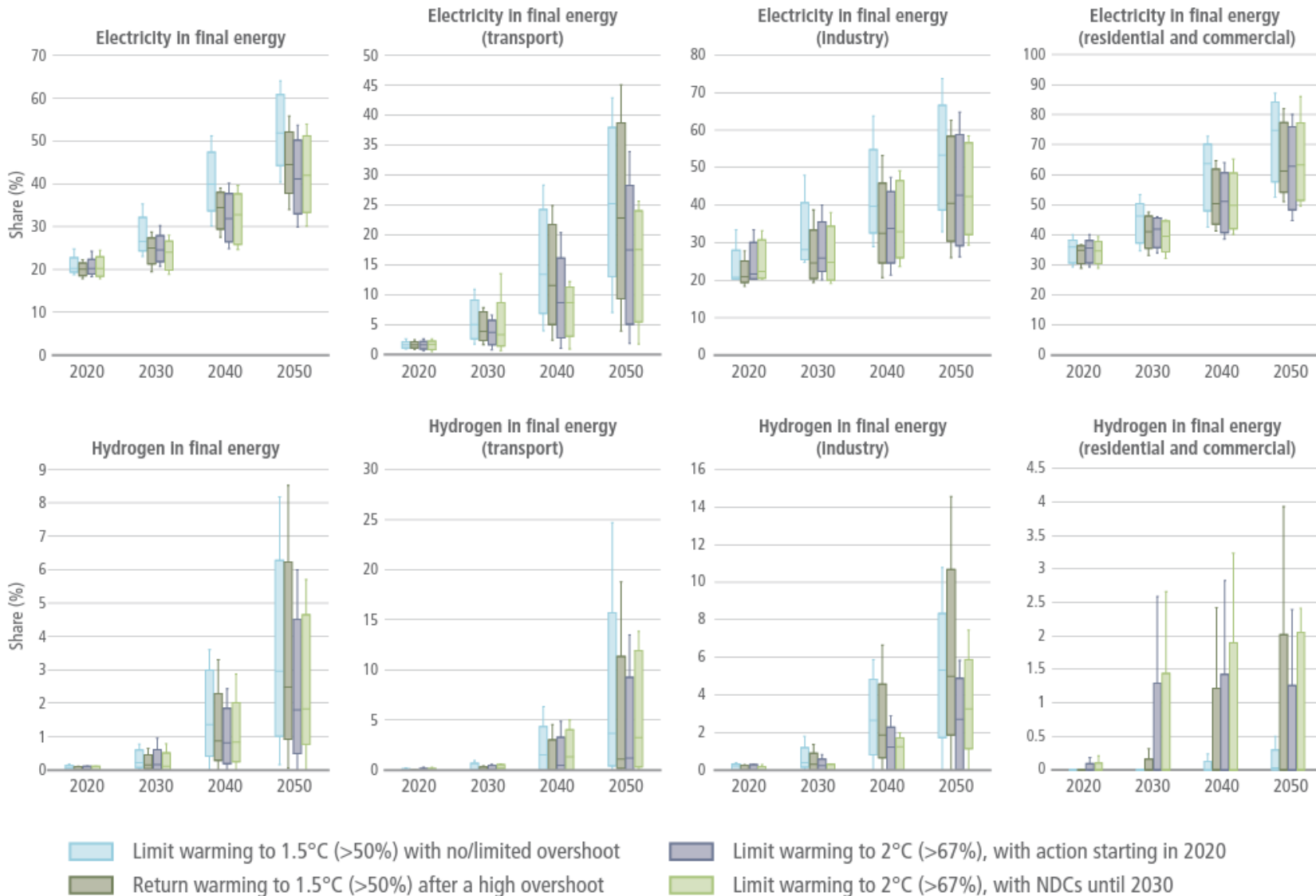
 Limit warming to 1.5°C (>50%) with no/limited overshoot
 Return warming to 1.5°C (>50%) after a high overshoot

 Limit warming to 2°C (>67%), with action starting in 2020
 Limit warming to 2°C (>67%), with NDCs until 2030

Scenarios consistent with Paris goals are characterized by decreases in fossil production and consumption



Scenarios consistent with Paris goals are characterized by increases in electricity and H2 in end-use sectors

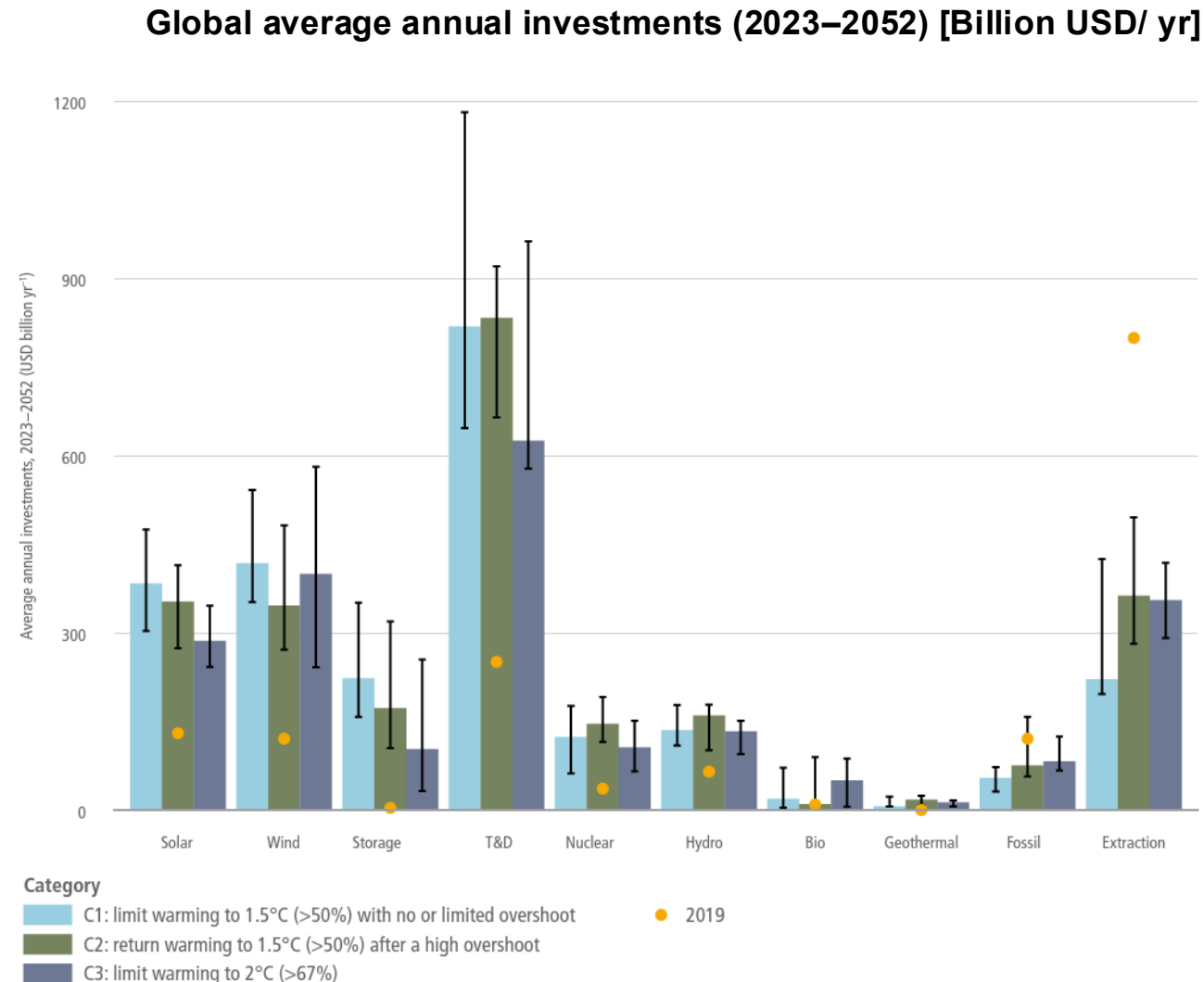


Boxes indicate 25th and 75th percentiles;

whiskers indicate 5th and 95th percentiles

Source:
IPCC,
AR6

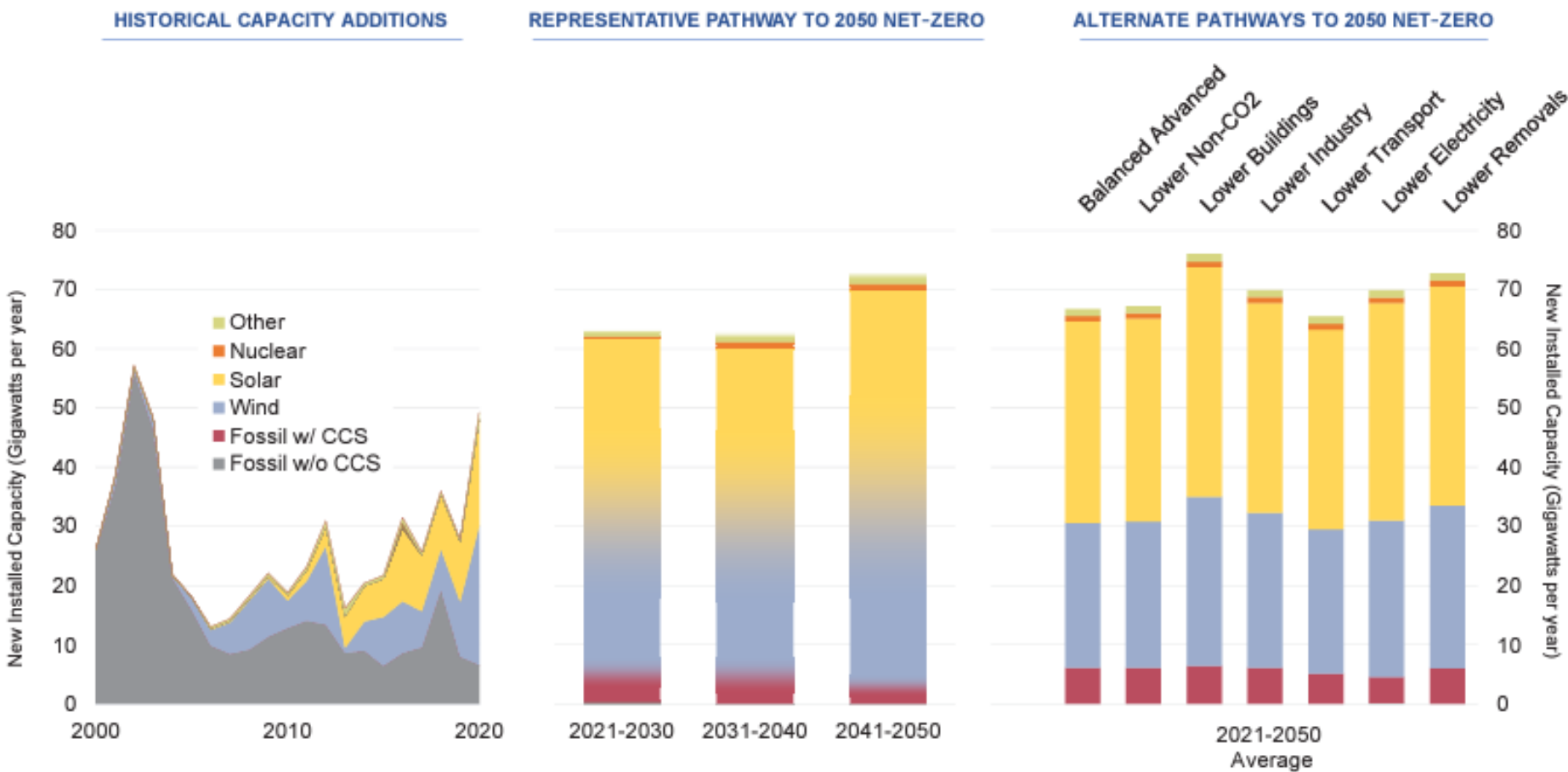
Scenarios consistent with Paris goals are characterized by changes in investment patterns



Bars show median values across models-scenarios, and whiskers the interquartile ranges

IAM based pathways toward net-zero emissions in the U.S. by 2050 suggest significant scaling up of investments compared to historical rates

U.S. Electric Generation Capacity Additions



Source:
U.S. LTS 2021

Scenarios consistent with Paris goals are characterized by stranded assets



<https://www.greenbiz.com/article/state-green-business-stranded-assets>

Stranded assets are “assets that suffer from unanticipated or premature write-downs, devaluations, or conversion to liabilities.”
(IPCC AR6 WGIII Glossary, Chapter 6, Box 6.1)

- ▶ In the context of climate change mitigation, stranded assets could manifest in various forms such as:
 - Fossil-fuel resources that cannot be burned in order to maintain a long-term temperature goal
 - Pre-mature retirement of carbon-intensive capital due to climate policies
- ▶ Mitigation could result in pre-mature write-downs, devaluations, and conversion to liabilities of carbon-intensive assets, referred to as “**stranding**” of assets.

Why are stranded assets important?

- ▶ Stranded assets could create financial and political threats:
 - **Economic risks:** Financial risks that could arise for parties that have suffered loss and damage from climate change mitigation.
 - **Political economy risks:** Stranded assets create loss of wealth for few and could result in lobbying and rent-seeking behavior.



<https://www.reminetwork.com/articles/climate-risk-raises-stranded-asset-potential/>

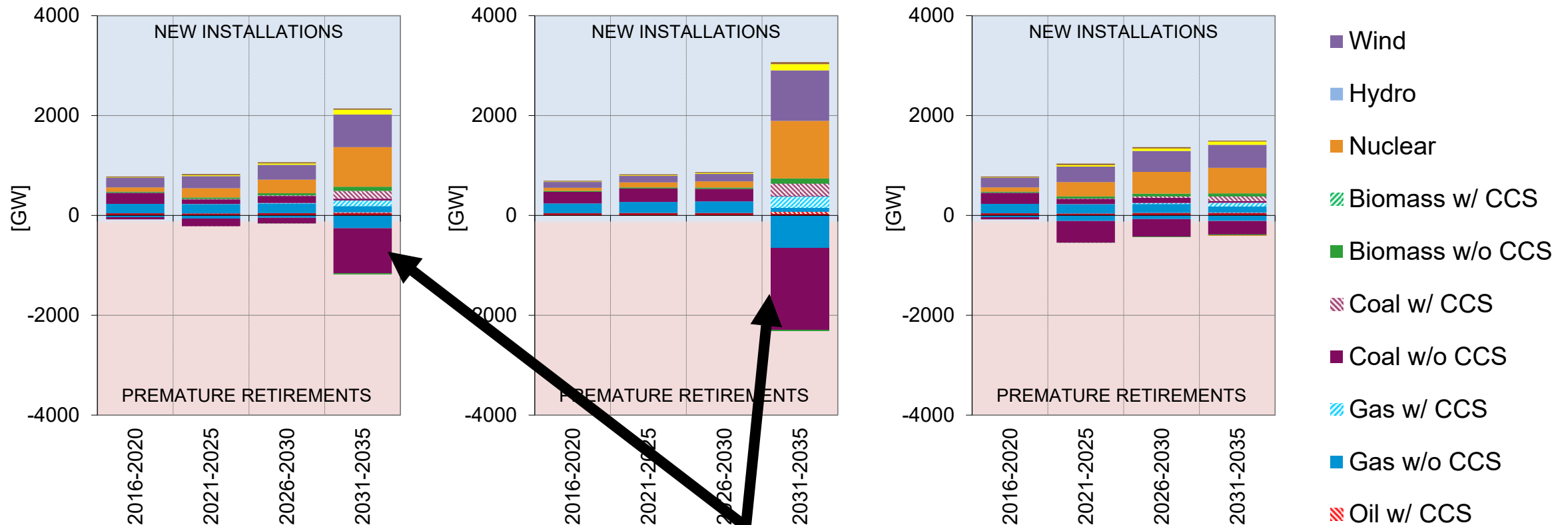
Globally, investments and stranded assets implications of Paris Agreement are non-trivial

Capacity Additions and pre-mature retirements [GW/yr]

Paris to 2°C

No Paris to 2°C

Least Cost 2°C



Current installed capacity globally is ~8000GW.
The *Paris*- and *No Paris* to 2°C scenarios suggest
pre-maturely retiring 20-25% of this capacity beyond 2030

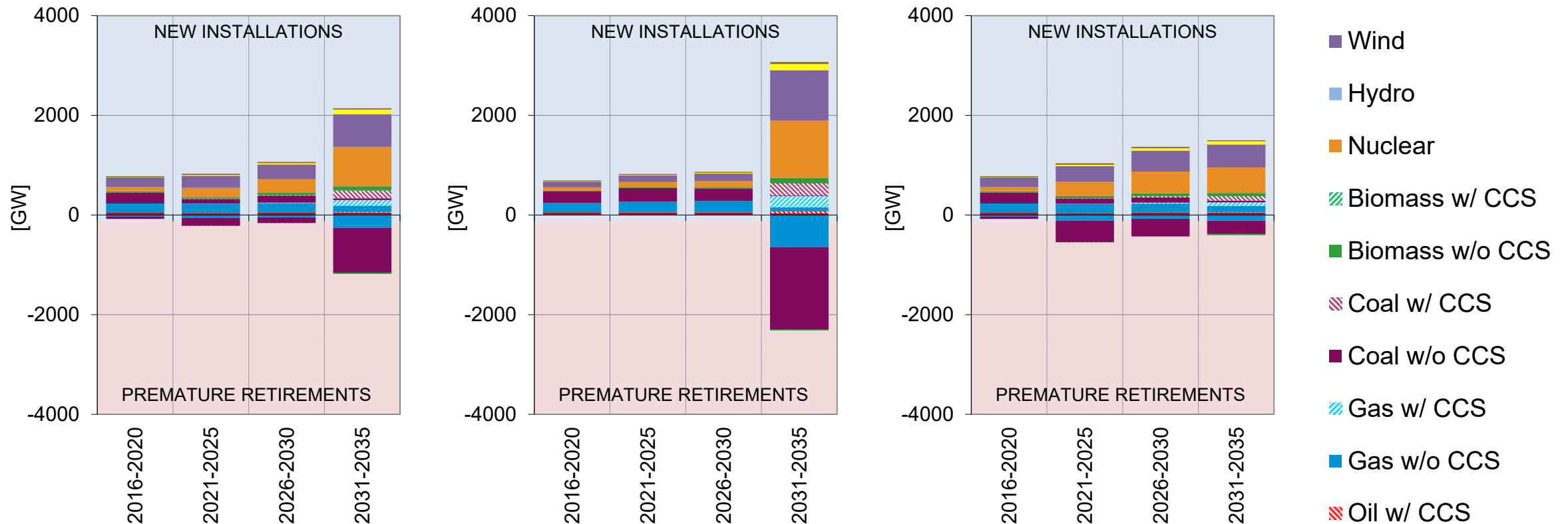
Globally, investments and stranded assets implications of Paris Agreement are non-trivial

Capacity Additions and pre-mature retirements [GW/yr]

Paris to 2°C

No Paris to 2°C

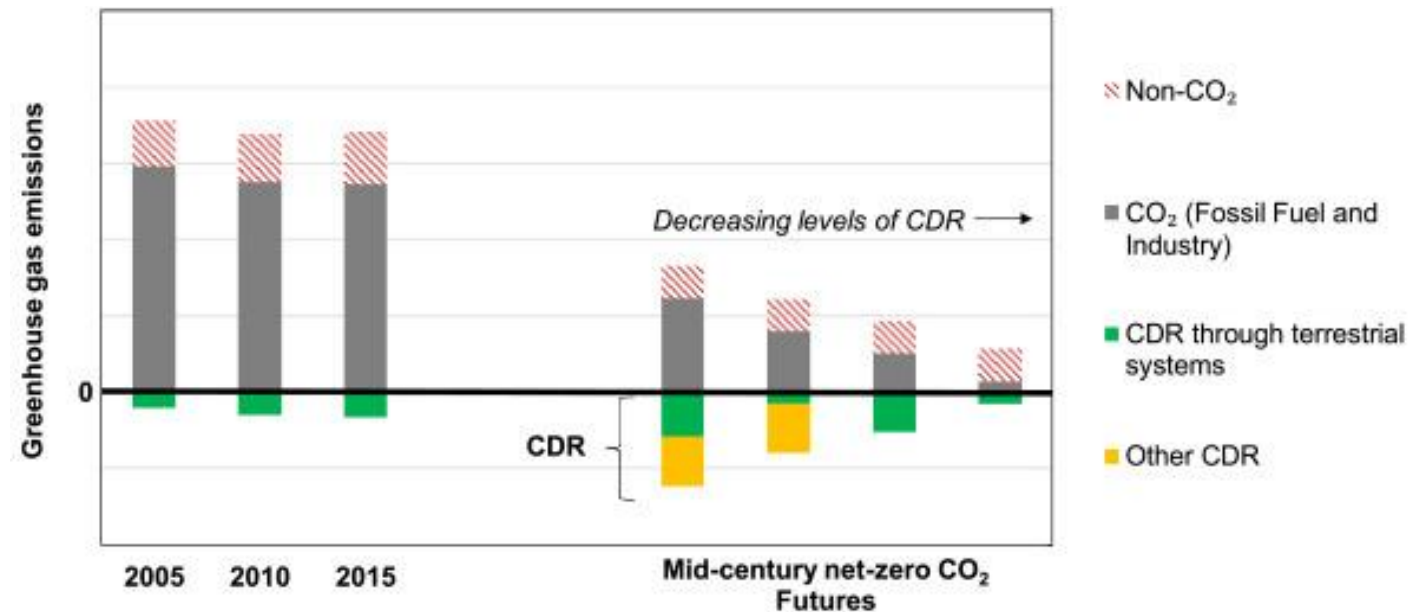
Least Cost 2°C



Stranded asset implications under 1.5°C would be even more prominent

Looking beyond the energy system: IAM-based research suggests that achieving economy-wide net-zero emissions is a zero-sum game

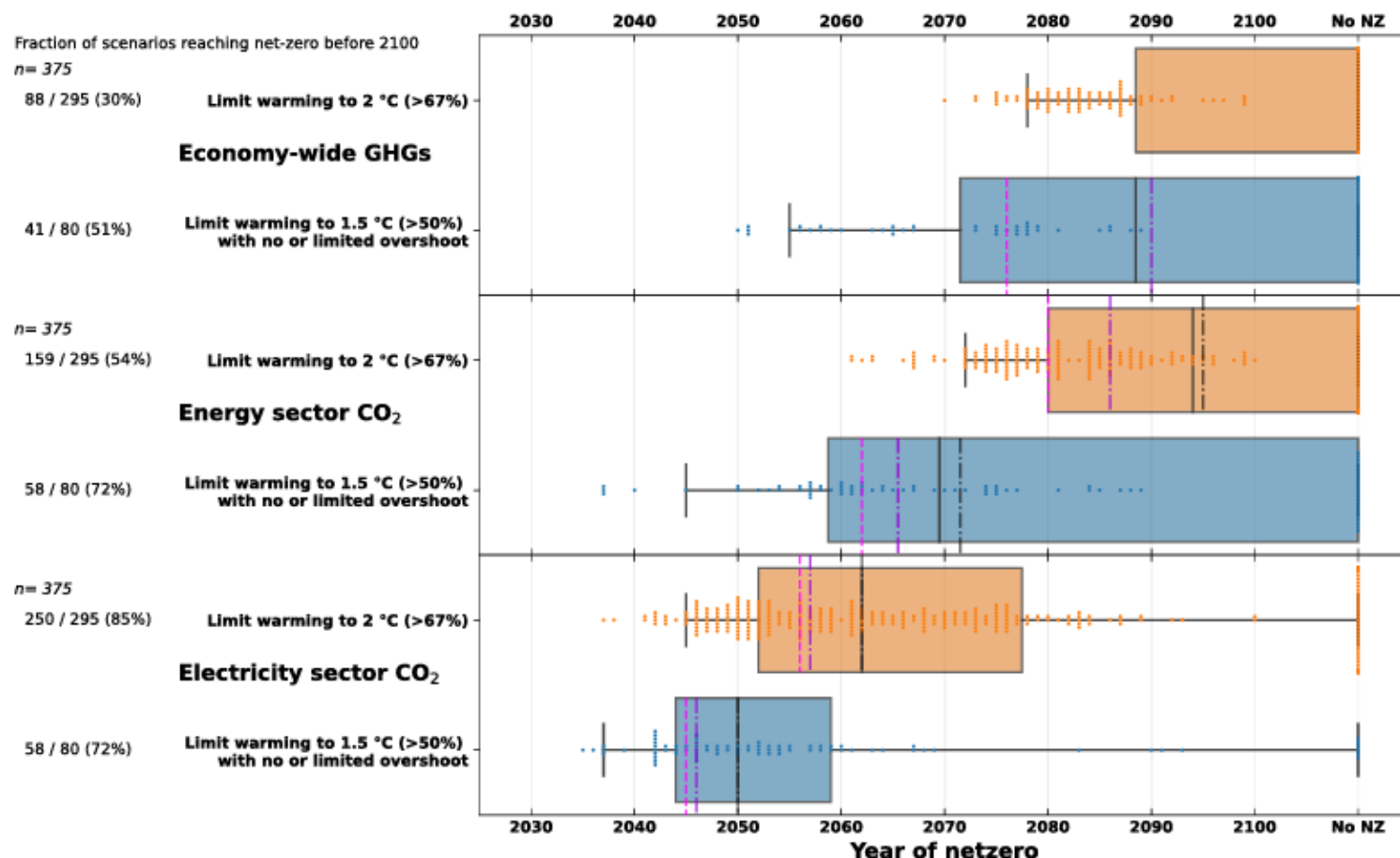
- Some non-CO₂ emissions are hard to reduce (e.g. from cattle)
- Carbon dioxide removal (CDR) will be critical to offset those emissions
- The greater the scale of CDR, the lower is the level of mitigation of CO₂ emissions from fossil fuels and industry and vice versa



Some emerging areas of energy systems research in IAMs

- Stranded assets under climate mitigation
- Effect of trade and geopolitical relationships on energy system transitions
- Linkages with mineral supply chains: Implications of mineral supply chains for energy system transitions and vice versa
- Linkages with sustainable development goals
- Spatial and sub-annual details to better capture intra-annual dynamics (e.g. storage in the power sector)
 - Impacts of changing intra-annual load patterns driven by climate, electric vehicle charging, etc.
- Impacts of behavioral changes
- Climate change impacts on hydropower, wind and solar
- Air Pollution impacts on solar generation

The timing of net-zero varies by sector



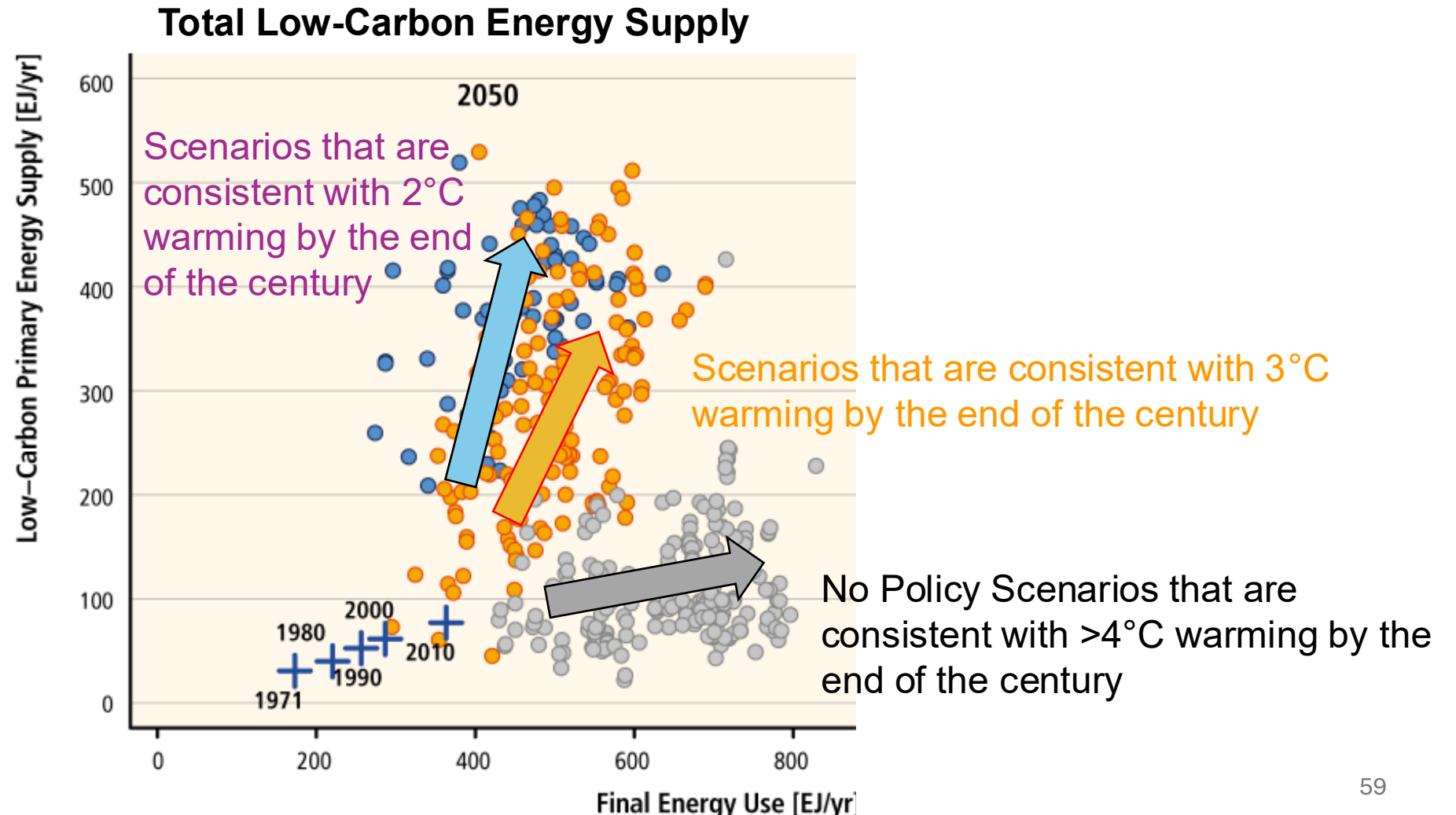
Black vertical line is the median across many IAM scenarios

Note that the range across scenarios is quite wide

Source: IPCC, AR6

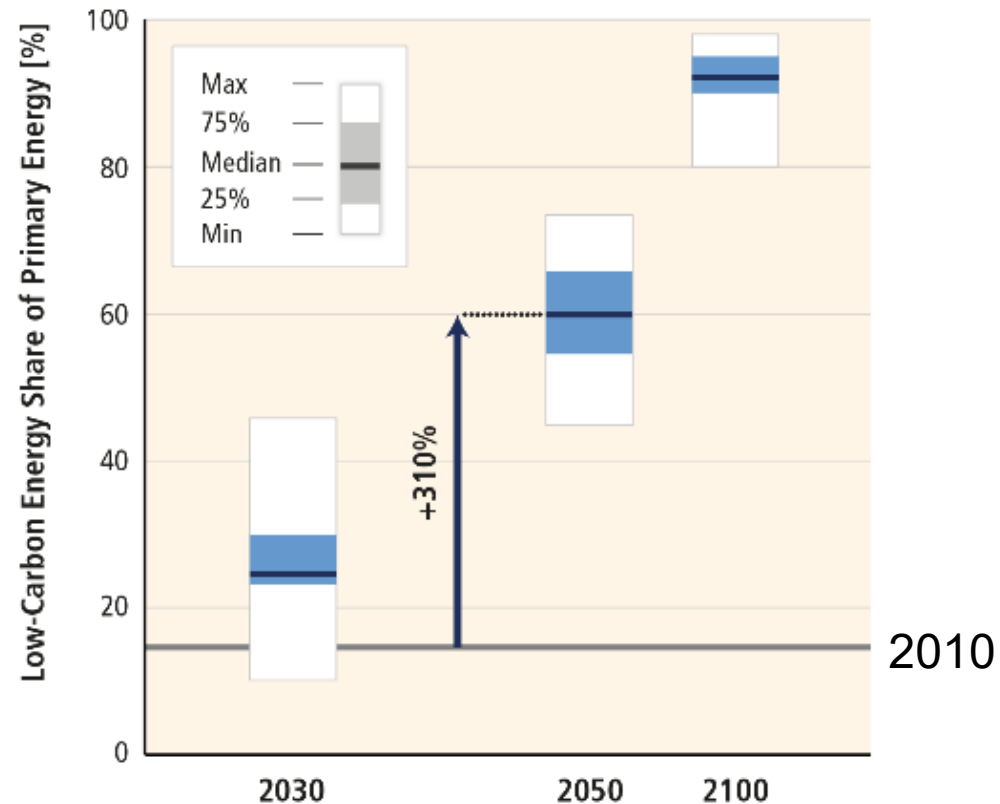
Boxes indicate 25th and 75th percentiles, **center black line is the median**, while whiskers indicate 1.5x the inter-quartile range. The vertical dashed lines represent the median point at which emissions in the scenarios have dropped by 95% (pink) and 97.5% (purple), respectively. Dots represent individual scenarios. The fraction indicates the number of scenarios reaching net-zero by 2100 out of the total sample. Source: AR6 Scenario Database.

Mitigation scenarios are characterized by a major upscaling of low- and zero- carbon energy



Mitigation scenarios consistent with 2°C warming are characterized by a tripling to nearly a quadrupling of the share of zero- and low-carbon energy supply

Scenarios that are consistent with 2°C warming by the end of the century



Source: IPCC, AR5

Thank you!

Muhammad Awais
awais.m@lums.edu.pk